ENVIRONMENT AND ECOLOGY

PAPER CODE-BAS104

B.TECH (FIRST SEMESTER)

CLASS NOTES (2023-24)

ELECTRONICS ENGINEERING

BY-ANURAG YADAV

UNIT-1: SYLLABUS

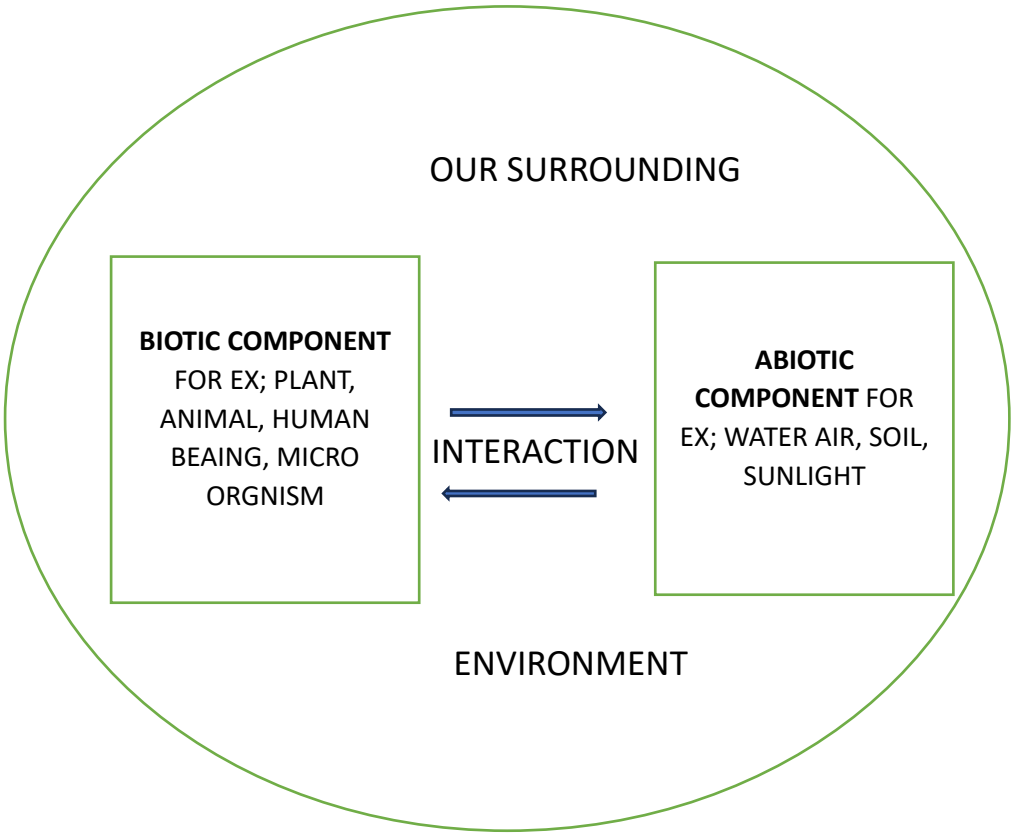
Environment: Definition, Types of Environments, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness.

Ecosystem: Definition, Types of ecosystems, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem.
Effects of Human Activities: Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development

ENVIRONMENT- DEFINITION; TYPES, COMPONENTS AND SEGMENTS OF ENVIRONMENT

Q: **What is Environment**?

Ans: The word environment has been taken from French word "**environner**" which means around or round about.



Definition of Environment: Environment can be defined in a number of ways-
◆ "Our surrounding which is formed by the interaction of living and non-living things, collectively known as our environment".

According to Mc. Naughton and Wolf-"Environment denotes the sum total of physical(=non-living) and biological (=living) factors that directly influence the survival, growth, development and reproduction of organisms."

What are the categories/types of Environments?

Ans:

Natural Environment: The environment that comes in its existence by its own or naturally without the influence of human beings is called as natural environment. It operates through a self-regulating mechanism that is known as Homeostasis. eg-Mountain, forest, lakes etc.

2. **Man-made or Anthropogenic Environment**:

The environment which has been modified by human activities is called man-made environment. When in a specific area, the temperature, humidity, etc., are controlled by human beings it is known as an anthropogenic environment. The increase in the scientific technology, which is the result of human brain, is deteriorating the natural environment. eg-water park, zoo etc.

What are the basic constituents (=components) of environment?

Ans: The environment of an organism has two basic components:

a**) BIOLOGICAL** (= Biotic)-for example: Plants, Animals, Human beings and Microorganisms.

B) **PHYSICAL** (-Abiotic)-for example: Air, water, soil, minerals, metals, rocks, rivers, fossil fuels, sunlight etc.

Q: What are the segments or parts of Environment
?

Ans: The environment consists of four segments:

1. **Atmosphere:** "Atmosphere is the gaseous cover around the earth surface."

Composition: Atmosphere comprises of air. Air is a mixture of gases. For example: Nitrogen (78.03%), Oxygen (20.99%), Carbon-Di-oxide (0.034%), inert gases (0.95%), water vapours and dust particles (variable).

On the basis of temperature variation, the atmosphere can be segmented as follows:

Troposphere
- Stratosphere
- Mesosphere
- Thermosphere (sub-divided into ionosphere and exosphere)

2. **Hydrosphere**: Hydrosphere comprises of water. More than 70% of the earth's surface is covered with water; therefore, earth is called blue planet. Of the total water available on the earth 97% water is salty and only 3% water is fresh.

salty water (97%) is present in: oceans, seas. estuaries

fresh water (3%)

2% water is in frozen form (snow ice form) and it is present in polar ice-caps and glaciers 1% water is available for human use it is present in rivers. lakes, ponds, wells and aquifers

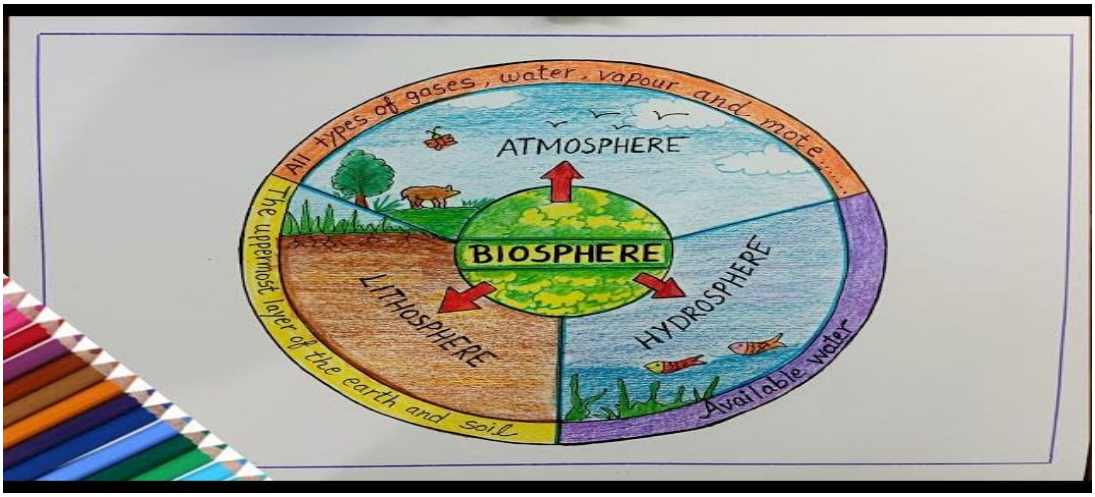
3 Lithosphere: Lithosphere comprises of soil, minerals and rocks.

✓ Soil is the upper fertile layer of earth which supports plants.

✓ Rocks are the hardest part of earth's crust. Rocks are of three types: - igneous, sedimentary and metamorphic.

4 Biosphere: Biosphere is the part of earth/environment which supports life. Biosphere is the sphere of living organisms (plant kingdom and animal kingdom). Biosphere extends to only 14 kilometres. This includes 6 kilometres in air and 8 kilometres in sea.

Biosphere = Atmosphere + Hydrosphere + Lithosphere and Life in them.



Describe the structure of atmosphere. How would you broadly divide major layers or region of the atmosphere? State respective altitude and temperature ranges.

Or

Classify atmosphere on the basis of temperature variation. State main activities taking place in each of them.

Or

Describe the atmospheric profile. Discuss different layers of atmosphere in detail.

Ans:

Atmosphere is a thick blanket (-envelope) of air which surrounds our earth. It extends to a height of 1600 kilometres from the earth's surface. The atmosphere is a reservoir of several elements essential to life.

Structure: -On the basis of temperature variation, the earth's atmosphere is divided into four different layers/zones.

i. Troposphere

ii. Stratosphere

iii. Mesosphere and

iv. Thermosphere (sub-divided into ionosphere and exosphere)

LAPSE RATE OR ENVIRONMENTAL LAPSE RATE: Change in atmospheric temperature with respect to height is called lapse rate.

a) Case1- If atmospheric temperature decreases with increase in height, it is called positive (+ve) lapse rate.

b) Case2- If atmospheric temperature increases with increase in height, it is called negative (-ve) lapse rate

Major regions of the atmosphere and their characteristics:

REGION	ALTITUDE (HEIGHT), Km	TEMPERATURE RANGE, °C	MAIN ACTIVITY	LAPSE RATE
Troposphere	0 to 18	17 to -55	Clouds formation, rainfall	-ive
Stratosphere	18 to 50	-55 to 0	O_3 formation	+ive
Mesosphere	50 to 80	0 to -75	Meteors burning	-ive
Thermosphere (Ionosphere ,exosphere)	80 to 640 and 640 to 10,000	-75 to more than 1200	Very high temperature	+ive

SCPOE AND IMPORTANCE OF ENVIRONMENTAL SCIENCE:

3. Environmental studies aware us about the environment, ecosystem and valuable natural resources, their importance and conservation methods.
4. Environmental studies provide knowledge about ecology & biodiversity.
5. -Environmental studies help us to solve the big environmental problems like Environmental Pollution, Urbanization, Deforestation, Desertification, Global Warming, Acid Rain, Ozone Layer Depletion etc.
6. Environmental studies help us to find ways to maintain ecological balance.

There are various fields/areas in which we can see the scope of environmental science.

a) Natural Resources, their conservation and management.

b) Ecology and Biodiversity conservation

c) Environmental pollution and its control

d) Global environmental issues and their control measures: Such as-Global warming, Ozone layer depletion, Acid rain, Desertification, Deforestation, Urbanization etc.

Q: Describe the need of public awareness about the environment.

Or

Discuss the need of public awareness for conservation and protection of environment.

Ans: The environment is degrading day by day because of increasing population, rapid industrialization, urbanization, deforestation and environmental pollution. All these problems have disturbed the ecological balance in such a way that the survival of life on earth has been seriously threatened. Therefore, public awareness is must for protection of environment.

• Public awareness can be achieved through environmental education i.e., people should be told about environment and its importance.

. Environmental education or environmental awareness among public can be spread through books, magazines (for ex: down to earth). schools! colleges/institutes/university, newspaper, media, television, radio, eco- clubs, Government and Non-Governmental Organizations (NGO'S).

Public participation is possible only when the public is aware about the environment and environmental issues (problems)"

Public can participate in/with:

1- In tree plantation

2- In Environmental movements for example:

Bishnoi Movement

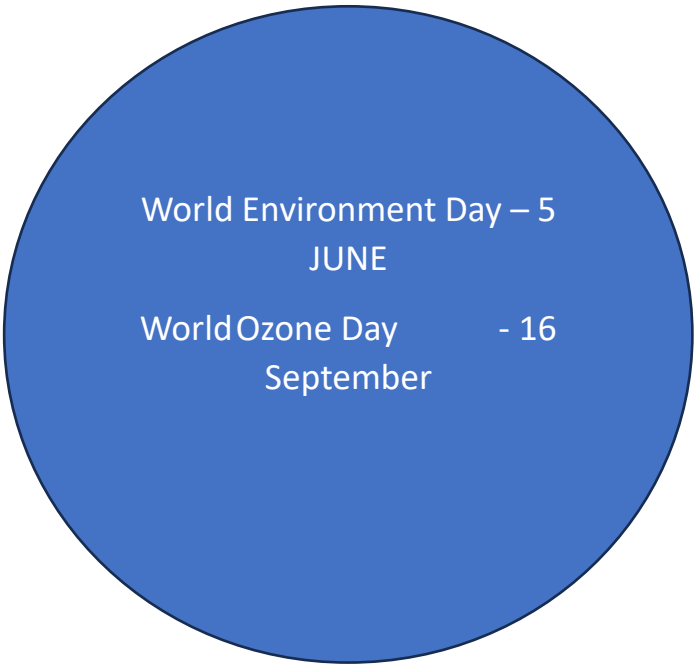
Chipko Movement,

Appiko Movement,

Silent Valley Movement,

3-With Eco-clubs/nature clubs

4-Public can work with Environmental NGO's



ECOSYSTEM

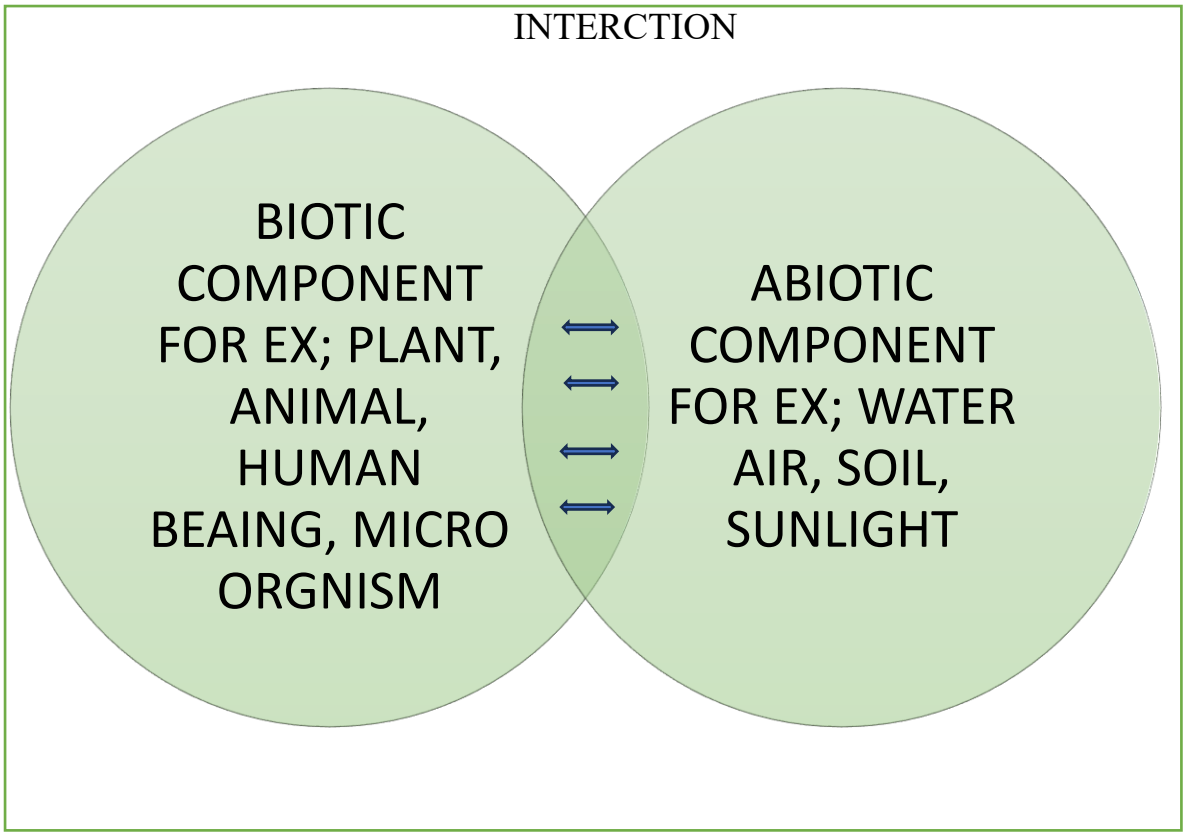
What is the concept of ecosystem?

OR Define the term ecosystem.

Ans: Definition of ecosystem:

The living (biotic) and non-living or physical (abiotic) components of the earth (environment) are linked and function (interaction) together as an ecological system or ecosystem. Therefore-

"The ecological system which is formed by the interaction of living (biotic) and non-living (abiotic) things is known as ecosystem". The term ecosystem was proposed by **A.G. Tansley (1935**).



Interaction In Systematic Way

Concept of ecosystem:

In an ecosystem, the interaction of life with its environment takes place at many levels. A single bacterium in the soil interacts with water, air around it within a

small space while a fish in a river interacts with water and other animals, rivals in a large space. Considering the operational point of view; the biotic and abiotic components of an ecosystem are so interlinked such that their separation from each other is practically difficult. So, in an ecosystem both organisms (biotic communities) and abiotic environment (rainfall, temperature, humidity) each influence the properties with other for maintenance of life.

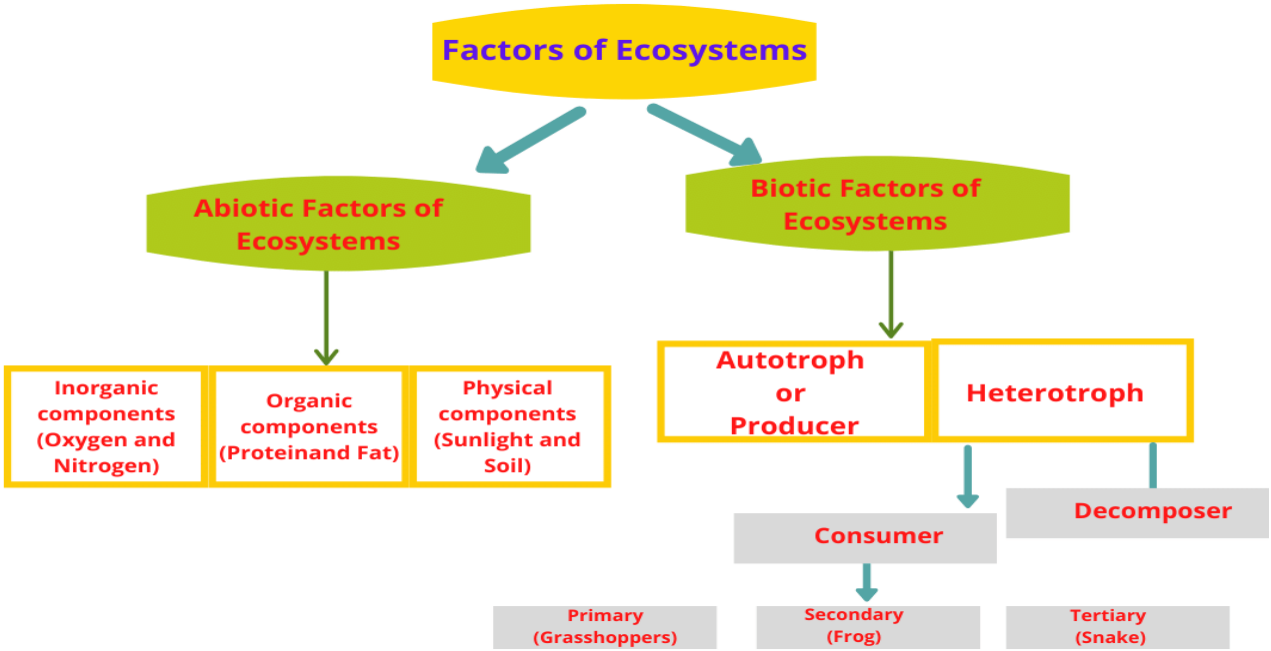
Ecology

The term ecology was given by **Ernst Haeckel in 1869.** Ecology has been defined variously

Ecology is the study of interaction (relationship) between organisms and their environment". OR

Ecology is the study of ecosystem OR Ecology is the study of environment.

STRUCTURE OF AN ECOSYSTEM:



There are two factors of an ecosystem; Biotic factor and Abiotic factor components.

Abiotic factor: Abiotic factor are the physical and chemical factors that directly or indirectly affect the living components e.g., air, water, land, rock etc. Non-living components are also called **Abiotic** components.

Physical components. include sunlight, water, fire, soil, air temperature etc.

Chemical factors include organic and inorganic component like- moisture, salinity of water, soil nutrients, oxygen dissolved in water

Biotic factor: Living components in an ecosystem are either producers or consumers.

They are also called **biotic** components. Producers can produce organic components e.g., plants can produce starch, carbohydrates, cellulose from a process called photosynthesis. Consumers are the components that are dependent on producers for their food e.g., human beings and animals•

Biotic factor is further classified into 3 main groups

•Producers •Consumers •Decomposers

1. **Producer (Autotrophs):** The green plants have chlorophyll with the help of which they trap solar energy and change it into chemical energy of carbohydrates using simple inorganic compound namely, water and carbon dioxide. This process is known as photosynthesis. The chemical energy stored by the producers is utilized partly by the producers for their own growth and survival and the remaining is stored in the plants for their future use. They are classified into two categories based on their source of food.

2. **Consumers (Heterotrophs):** The animals lack chlorophyll and are unable to synthesis their own food therefore they depend on the producers for their food. •They are known as heterotrophs (i.e., heteros= others, trophs= feeder). The Consumers are of 4 types:

(a) Primary Consumer: (Herbivores) i.e., Animal feeding on plants, e.g., Rabbit, deer, goat etc.

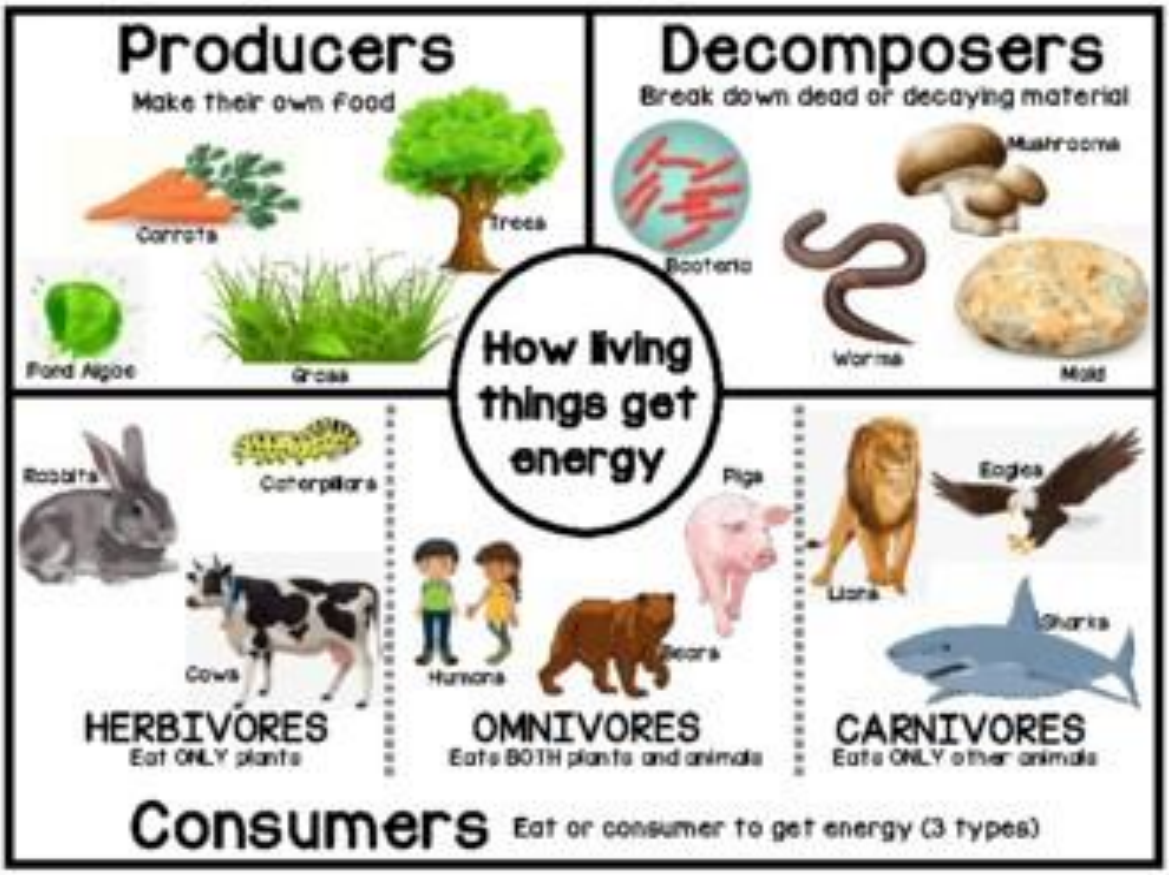
(b) Secondary Consumers: The animal feeding on Herbivores is called as secondary Consumer or primary carnivores. e.g., Cats, foxes, snakes.

(c) Tertiary Consumers: These are large carnivores which feed on secondary consumers. e.g., Wolves, Eagle

(d) Quaternary Consumers: They are also called omnivores these are largest carnivores Which feed on tertiary consumers and are not eaten up by any other animals. e.g., lion and Tiger.

3. **Decomposers**: Bacteria & fungi belong to this category. They break down the dead organic matter of producers & consumers for their food and release to the environment the simple inorganic and organic substance. These simple substances are reused by the producers resulting in a cyclic exchange of material

between biotic & abiotic environment. **Eg: Bacteria, Earth worms,Beetles,etc**



CLASSIFICATION OF ECOSYSTEMS

Due to the abiotic factors, different ecosystems develop in different ways. These factors and their interaction between each other and with biotic components have resulted in formation of different types of ecosystems as explained below. Ecosystem may be natural or artificial.

. **I) Natural Ecosystems:** These ecosystems operate by themselves under natural conditions without any major interference by man. Based upon the particular kind of habitat, these are further divided as: •Terrestrial as forest, grassland, desert etc.

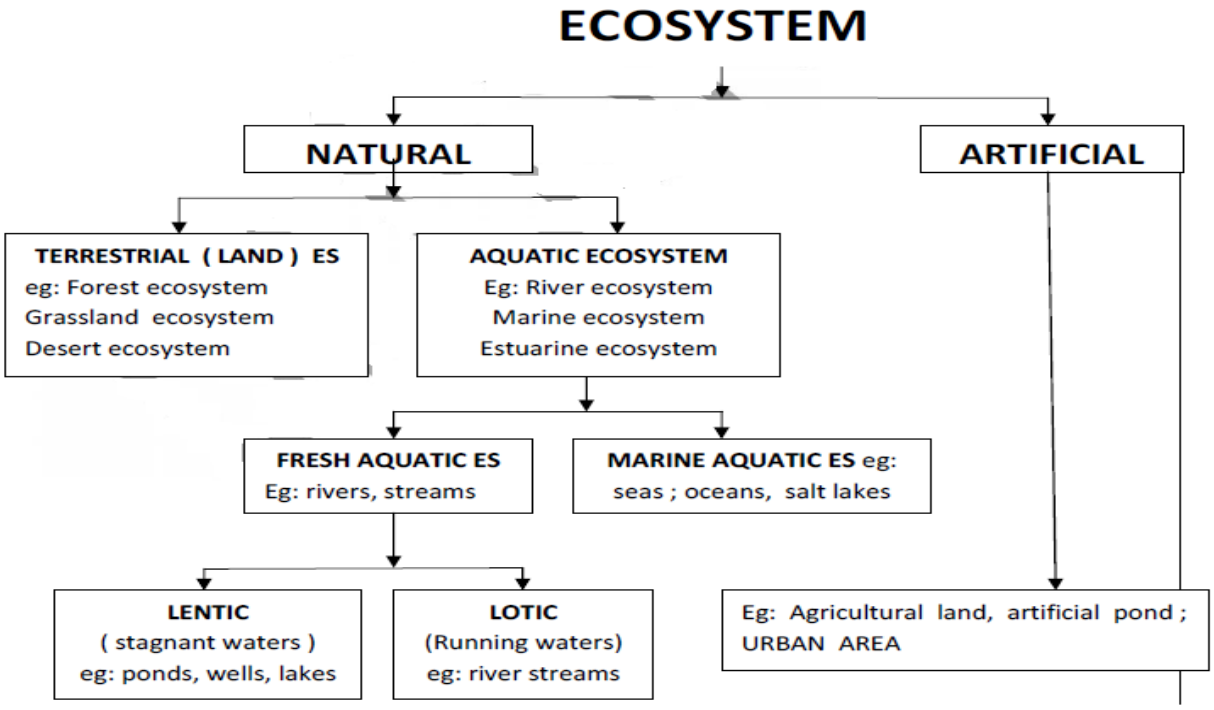
•Aquatic which may be further distinguished as

•Freshwater which may be lotic (running water as springs, stream, river) or lentic (standing water as lake, pond, pools, ditch, swamps, etc.)

•Marine Ecosystems: as an ocean or shallow ones like sea or estuary etc.

II) Artificial Ecosystems: These are maintained by man where, by addition of energy & planned manipulations natural balance is disturbed regularly.

For eg : croplands like maize, wheat, rice-fields etc., where man tries to control the biotic community as well as physio-chemical environment are artificial ecosystems



Describe the salient features, structure and functions of forest ecosystem.

ANS: Forests occupy 23-40% of the land area in the world. In India about 17-22% of the total land area is under forests. Depending upon the climatic conditions, the forests have been divided into four categories:

- Tropical forests
- Sub-tropical forests
- Temperate forests and
- Alpine forests

The chief components of a forest ecosystem are as follows:-

Abiotic component	**Biotic component**				
	Producers	**Herbivore or Primary consumers**	**Secondary consumer OR Carnivore 1**	**Tertiary consumer OR Carnivore 2**	**Decomposer**
Soil, humus, water, sunlight, air	Mainly Trees Herbs, Shrubs and Grass	Mouse, Rabbit, Deer, Sheep, Cow	Lizards, Birds Snakes, Fox	Lion, Tiger, Wild Cat	Bacteria and Fungi

Functions of FORESTS:

Productive functions	**Protective functions**	**Regulative functions**
• We get a lot of valuable products from forests like: • Fruits • Fodder • Fuel wood • Fibres • Wood for furniture • Rubber • Gums • Honey • Medicines etc.	• They provide protection to wild animals • They provide protection against Soil erosion, drought, flood and landslide. • They help to conserved soil and water.	• Regulates atmospheric temperature • Regulates water cycle • Regulates energy cycle • Regulates nutrient cycling (C,H,O,N,P,S cycle) • They purify air (atmosphere) • They control environmental pollution

Write a short note on Grassland ecosystem.

Ans:

Grassland is an ecosystem in which grasses and other forage plants are the dominant vegetation. Grasslands occupy about 19% of the earth's surface. Grasslands are present where there is not sufficient rainfall to support large trees. The major grassland ecosystems of world are:

Great Plains of Canada and United States.

South Argentina to Brazil

South Asia to Central Asia.

The various components of a grassland ecosystem are as follows:

Abiotic component	**Biotic component**				
	Producers	**Herbivore or Primary consumers**	**Secondary consumer OR Carnivore 1**	**Tertiary consumer OR Carnivore 2**	**Decomposer**
Soll, Water, Sunlight, Air	Grasses, Herbs and Shrubs	Grasshopper, Rabbit, Mouse, Deer, Sheep Etc.	Frog, Snake, Lizards, Birds, Fox, Jackal Etc.	Hawk/ Eagle and Vultures Etc.	Bacteria and Fungi

Write a short note on desert ecosystem. Ans:

Desert ecosystem occurs in the region where annual rainfall is less than 25 cm.

They occupy about 17% of the land area.

Due to high temperature and scarcity of water the biota (plants, animals and micro-organisms) is less.

Abiotic component	**Biotic component**				
	Producers	**Herbivore or Primary consumers**	**Secondary consumer OR Carnivore 1**	**Tertiary consumer OR Carnivore 2**	**Decomposer**
High temperature, less rainfall, less water and poor nutrients Soil	Shrubs, Cactus	Camel, Kangaroo rat, Rabbits	Reptiles, Snakes.	Vultures	Thermophilic bacteria and fungi

Explain the structure of pond/lake ecosystem.

Ans:

Pond is a small fresh water aquatic ecosystem where water is stagnant (standing). The fresh water pond as whole represents a complete self-maintaining and self-regulating ecosystem. The various components of a pond/lake ecosystem are as follows:

Abiotic component	**Biotic component**				
	Producers	**Herbivore or Primary consumers**	**Secondary consumer OR Carnivore 1**	**Tertiary consumer OR Carnivore 2**	**Decomposer**
Water, mud, dissolved gases and nutrients in water, sunlight, air etc.	Phytoplankton's (Algae) and Aquatic plants	Zooplanktons (Krill, and jellyfish.)	small fish	Large fish	bacteria and fungi

CONCEPT OF FOOD CHAIN, FOOD WEB AND ENERGY FLOW

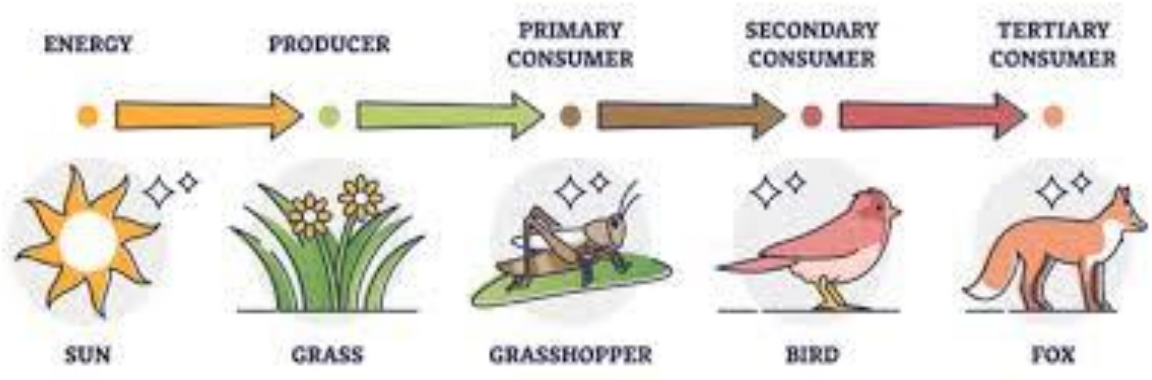
Q: What is food chain? Give its types

Ans: The food energy produced by green plants (producers) is consumed by herbivores and herbivores in turn are consumed by carnivores. This sequential process of eating and being eaten is called food chain. Definition: "The transfer of food energy from organism to organism i.e., from producers to consumers to decomposers in an ecosystem is known as food chain".

Examples-1**:** FOOD CHAIN OF A FOREST ECOSYSTEM:

HERBS, GRASS, TREE **(PRODUCERS)**	→	DEER **(HERBIVORE)**	→	LION **(CARNOVORE)**

Examples-2**:** FOOD CHAIN OF A GRASSLAND ECOSYSTEM:



TYPES OF FOOD CHAIN:

Food chains are mainly of two types:

1- **Grazing food chain**: it starts from producers and ends up at carnivores.

EX:

HERBS, GRASS, TREE **(PRODUCERS)**	→	DEER **(HERBIVORE)**	→	LION **(CARNOVORE)**

2. **Detritus food chain**; it starts from detritus and ends up at detritus feeder.

This type of food chain starts from dead organic matter (detritus) and goes to primary detritus feeder and then on to secondary detritus feeders.

For example:

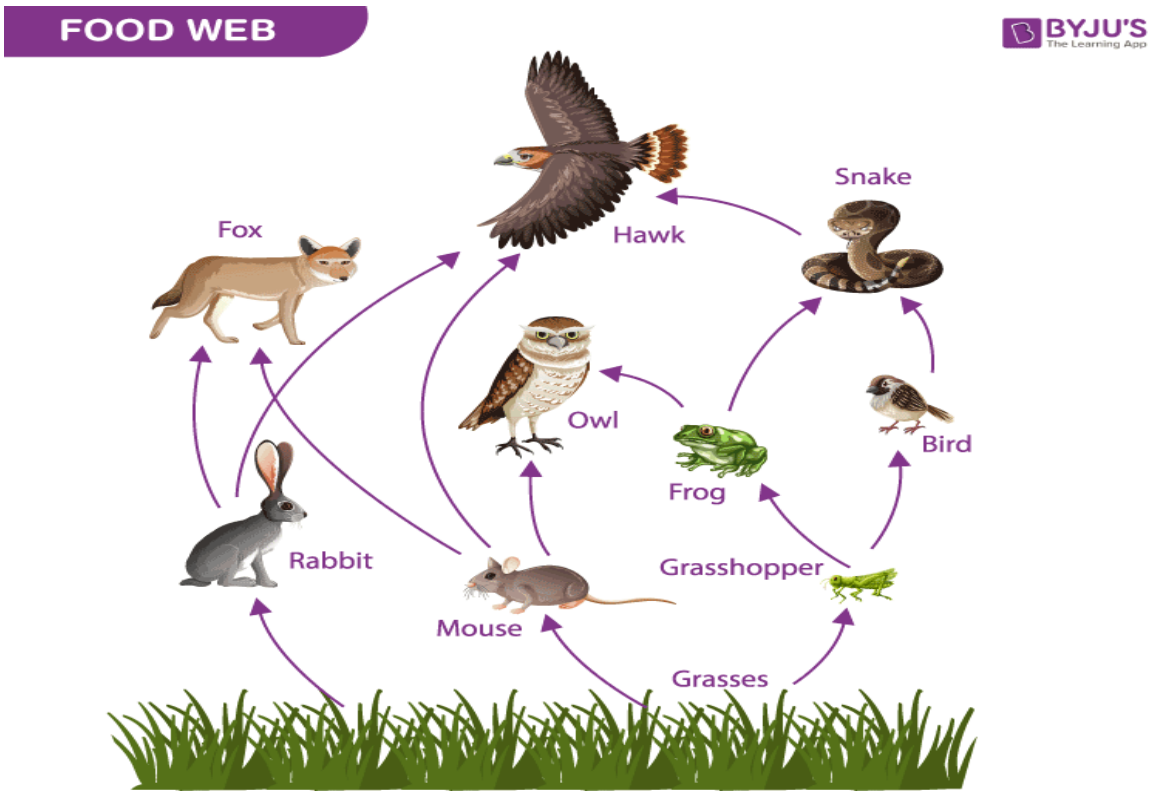
Litter → Earthworms → Chicken → Hawk

Detritus food chain

Q: **What is food web?**

The existence of single food chain is not possible in any ecosystem, Food chains are not always simple and isolated but are interconnected with one another. therefore, several food chains are combined (-linked together) to form a network which is called food web.

For example: In a grassland ecosystem grasses and plants are eaten by a number of animals such as rabbit, rat (mouse), grasshopper etc. These animals are eaten by snakes, frogs and lizards which in turn are eaten by eagle or hawk. Thus a complex network of food chains is formed which is called food web.



CONCEPT OF TROPHIC LEVELS.

The concept of trophic level was first given by Lindeman (1942).

According to him- a food chain consists of steps/levels/links. These steps or links are called trophic levels or food levels.

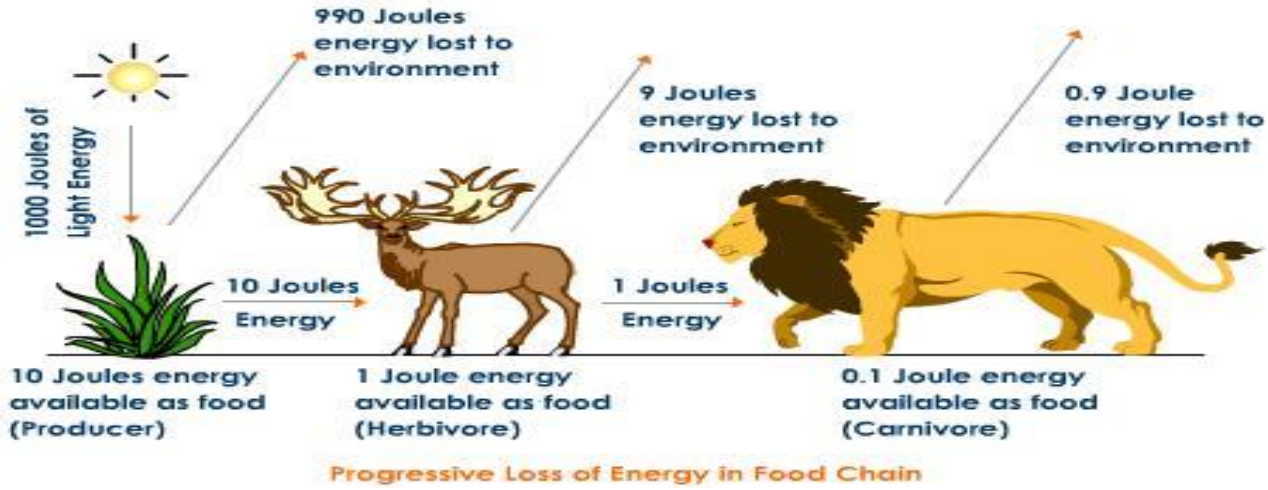
Type of organism	**Trophic level**
Producers	First trophic level (T_1)
Herbivores Or Primary Consumers	Second trophic level (T_2)
Secondary Consumers Or Carnivore 1	Third trophic level (T_3)
Tertiary Consumers Or Carnivore2	Forth trophic level (T_4)
Quaternary Consumers Or Carnivore3…..	Fifth trophic level (T_5)

Example: Food chain of a forest ecosystem with 3 Trophic Levels:-T1, T2, T3

HERBS, GRASS, TREE → DEER → LION

($\mathbf{T_1}$) ($\mathbf{T_2}$) ($\mathbf{T_3}$)

10% ENERGY LAW: The ten percent energy law was given by Lindeman (1942). According to this law "The energy available at a particular trophic level is 10% of its previous trophic level"



ECOLOGICAL PYRAMIDS AND THEIR TYPES

Ecological pyramids are the graphical representation of various trophic levels of the food chains in an ecosystem.

Types of pyramids:

1-Pyramid of number

2-Pyramid of biomass

3-Pyramid of energy

Pyramid of number: "It is the graphical representation of number of organisms at each trophic level of the food chain in an ecosystem, The pyramid of number may be upright or inverted,

Example 1: **Pyramid of number**

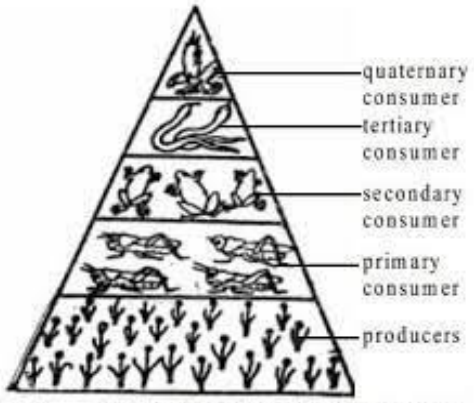
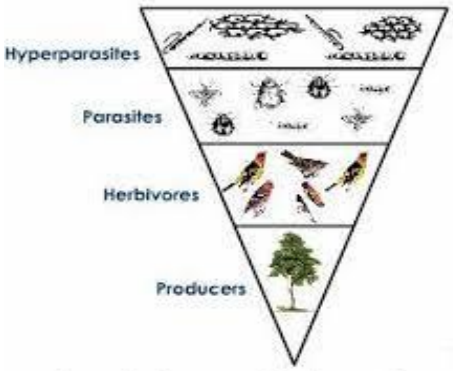
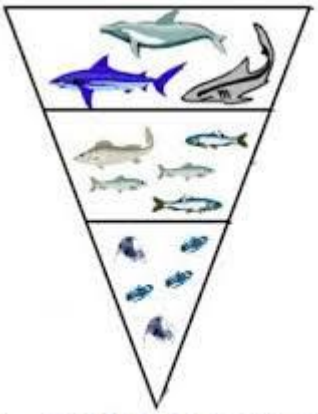


Fig. Pyramid of numbers in a grassland



Inverted pyramid of number

2. **Pyramid of biomass**: “it is the graphical representation of amount of biomass (=dry mass) available at each trophic level in an ecosystem". The pyramid of biomass may be upright or inverted.



Pyramid of Biomass : Ocean Ecosystem

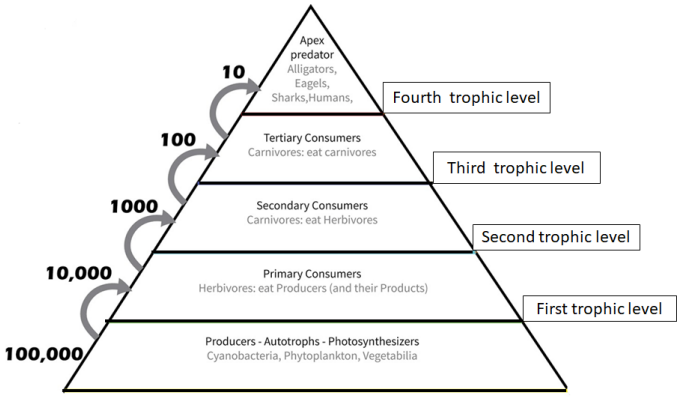
Pyramid of energy:

"it is the graphical representation of amount of energy content available at each trophic level in an ecosystem”

The pyramid of energy is always upright (i.e. the energy pyramid can not be inverted in a stable ecosystem) because there is a gradual decline in the amount of energy as we move from producers to consumers (as per 10% energy law).

✓ The energy available is maximum at producer level

✓ The energy available is minimum at consumer level



Pyramid of energy

HOMEOSTASIS OR BALANCE IN ECOSYSTEM

Homeostasis is a tendency of an ecosystem to resist changes so as to remain in a state of greater stability (= or equilibrium or balance).

A state of balance between living and non-living components of ecosystem is called ecological balance. Balance in an ecosystem is maintained by energy flow and nutrient cycling. organisms of ecosystem. A balanced ecosystem is essential for the survival of all

If producers are destroyed in nature, the primary consumers (herbivores) would die due to overcrowding and competition and lack of food.

If decomposers are completely destroyed from an ecosystem, the ecosystem functioning will be adversely affected because the mineral movement will be blocked.

HUMAN ACTIVITIES:

People residing at both rural and urban areas perform various activities to carry on their lives. For example: -

- Agricultural & Allied Activities,
- Mining & Allied Activities,
- Industrialization
- Technological development
- Irrigation projects
- Road & building construction
- Housing, Urbanization & Transportation,

IMPACT OF HUMAN ACTIVITIES ON ENVIRONMENT

- Environmental degradation
- Environmental pollution (i.e., air pollution, water pollution, marine pollution,
- Pollution (i.e. soil pollution, radiation pollution, noise pollution, thermal pollution etc.)
- Pollution induced problems like global warming, ozone layer depletion, acid rain, eutrophication, biomagnification etc.
- Resource crunch
- Depletion of natural resources (i.e., depletion of air, water, soil, minerals, energy and forest resources)

- Loss of wildlife
- Desertification
- Defacing landscape
- Soil erosion
- Hydrological deterioration
- Global climate change
- Ecological imbalance

EIA AND ITS PROCEDURE

EIA (Environmental Impact Assessment) is a systematic process of assessment of impacts of any development project (for ex: factory/industry/dam etc.) on environment.

- EIA is one of the recent the branch of environmental engineering.
- In India, EIA is conducted by MoEF (Ministry Of Environment And Forest).

Aims/objectives of EIA:

- To achieve sustainable development.
- Minimum (minimal) environmental degradation.
- Prevention from long term adverse impacts.

STEPS/STAGES/METHODOLOGIES INVOLVED IN EIA PROCESS:

These are as follows:

1-BASE LINE STUDY OR INITIAL SCRUTINY: it involves-

- Demarcation of study area
- Initial survey or first real visit of site
- Study of project details and environmental quality

2- **IDENTIFICATION OF PROJECT IMPACTS**:

- Air
- Water
- Land
- Plants

- Animals
- Human Beings
- Society
- Economics

MEASUREMENT OF MAGNITUDE OF IMPACTS:

Magnitude may be

- Nil
- Low
- Moderate
- High

AGGREGATION OF PROJECT IMPACTS

- Neutrals
- Positive
- Negative

ENVIRONMENTAL IMPACT STATEMENT

EIS (Environmental Impact Statement) is a document written in the format as specified by Department of Environment. EIS is a part of EIA process and it consists of data regarding the impacts of development project on environment.

It includes:

- Impact of project on land quality.
- Impact of project on water quality.
- Impact of project on air quality.
- The extent of noise produced.
- Impact of project on plants and animals (i.e. biodiversity).

- Impact of project on National Parks, Wildlife Sanctuaries.
- Impact of project on society.
- Impact of project on public health.
- Impact of project on agriculture and tourism

REVIEW AND DECISION MAKING: The decision maker can:

1. Accept the project
2. Reject the project
3. Accept an alternative
4. Request for further review
5. Post auditing or monitoring

FLOW CHART OF EIA

Discuss the concept of sustainable development.

Sustainable development is:

- An eco-friendly development
- A long-term development
- Such a development which helps in maintaining/sustaining the availability of natural resources for present generation as well as future generation without affecting their quality.

WAYS TO ACHIEVE SUSTAINABLE DEVELOPMENT:

Sustainable development can be achieved in various ways:

1. By checking population explosion

2. By implementing 4 "R" policy i.e.

- Reduce
- Reuse
- Recycle
- Recovery

3. By using appropriate technology (=eco-friendly technology)

4. By judicious use of natural resources.

5.Through environmental awareness, environmental education and public participation.

6.By Implementing laws (Acts) and legislation.

OBSTACLES IN THE PATH OF SUSTAINABLE DEVELOPMENT:

- Pollution
- Over-Population
- Poverty
- Discrimination (=inequity) against women
- Caste based fight
- Religion based fights
- Dirty politics
- Problems of female infanticide, child marriage, child labour and dowry
- Illiteracy (lack of education)
- Corruption

BASIC OBJECTIVES OF SUSTAINABLE DEVELOPMENT.

i. Firstly, it satisfies the basic needs of people regarding food, water, air, shelter for present and future.

ii. Secondly, these basic needs are fulfilled without depleting or degrading the earth's natural resources so that future generations may also.

NEED FOR SUSTAINABLE DEVELOPMENT

The developmental activities are putting an a lot of unnecessary (undue) pressure on the natural resources like air, water, soil, plants and animals. If all this goes unchecked, the day is not far off when there will be a condition of resource crisis (resource crunch).

The need for sustainable development can be summarised by following points:

- To meet the needs of present as well as future generations.
- To utilize natural resources judiciously (with care) so that their continuous supply can be maintained in the future.
- To undertake (carry out) developmental activities keeping in view the stability of ecosystem.
- To protect the world's biodiversity for future generations.
- To develop less polluting technologies
- To develop a strong economy for the growth of nation and its people because the economic growth of a country depends upon the availability of natural resources.

AKTU: B.TECH. SYLLABUS

Unit-2
Natural Resources: Introduction, Classification.
Water Resources; Availability, sources and Quality Aspects, Water Borne and Water Induced Diseases, Fluoride and Arsenic Problems in Drinking Water. Mineral Resources; Material Cycles; Carbon, Nitrogen and Sulphur cycles. Energy Resources; Conventional and Non-conventional Sources of Energy. Forest Resources; Availability, Depletion of Forests, Environment impact of forest depletion on society.

What are Natural Resources: The resources which have been provided by the nature for the survival of living beings are known as natural resources. For ex: Air, water, soil, sunlight, plants (forests), animals, minerals, metals, fossil fuels, nuclear fuels etc.

Classification of Natural Resources:

Depending on the basis of their abundance and availability, natural resources are of following two types-

A. Inexhaustible resources:

These resources are unlimited or unending and are not likely to be exhausted (finished) by human activities. These resources naturally perpetuate (continuously forming) themselves. Human activities can alter (change) the quality of these resources but not their quantity. For example: Sunlight, wind, water (rainfall), tides etc.

B. Exhaustible resources: These resources are limited. Both quantity and quality of these resources can be altered by human activities. These resources are of two types-

i. Renewable resources:

The resources which can be renewed (=regenerated=recycled) by nature or human activities after a time interval are known as renewable resources. For ex: Plants (forests), animals, soil, and groundwater etc.

ii. Non-renewable resources:

These resources are formed in nature over millions of years and which once exhausted, can not be recycled or regenerated in short time. For ex: Fossil fuels (Coal, Petroleum, and natural gas), minerals, metals and nuclear fuels etc.

WATER RESOURCES (DISTRIBUTION, SOURCES AND QUALITY ASPECTS)

Water is also called lifesaving liquid .Water is the major constituent of the hydrosphere. The total water in hydrosphere is 1.4 billion cubic kilometres (km^3).

✓ More than 70% of the earth's surface is covered with water; therefore earth is called blue planet. the total water available on the earth, 97% water is present in oceans and seas. It is salty or saline in nature and hence it is unsuitable for human consumption. And only 3% water is fresh water. Of this 3% of fresh water, 2% water is contained in polar ice-caps and glaciers. Due to their location this form of water is inaccessible (i.e. difficult to get). And remaining 1% of the fresh water is easily available and suitable for human use. It is found as underground water, as surface water (i.e. in rivers, lakes, ponds etc.) and as water vapours in the atmosphere. In nature there exists a continuous circulation of water between earth and the atmosphere; which is known as water cycle or hydrological cycle

Sources of water: Water resources can be classified into two types i.e.

1) **Salt water resources**: For example-ocean and seas

2) **Fresh water resources**: For example- rainfall, surface water (ponds, lakes, rivers, springs, water streams), ground water.

Uses of water: Water is used variously. For example-

1. **Domestic activities**: Bathing, cooking, cleaning, washing, drinking, and for cattles.

2. **Institutional:** Hotels, restaurants, offices, educational institutions.

3. **Agricultura**l: For irrigation (maximum consumption of water occurs in irrigation)

4**. Industries:** as raw material, for boilers, for cooling condensers etc.

5. Fire fighting

6. For electricity generation

7. Transportation

8**. Photosynthesis**: As raw material

9. **Absorption**: Mineral nutrients get dissolved in water and are absorbed by plants.

10. Disposal of sewage and other organic waste

WATER CONSERVATION.

Water conservation means utilization of water in a sustainable manner so that it could fulfil present as well as future demand. Followings are the measures for conservation and management of water:

1. **Reduction in consumption of water**: -

- Reduction of wastage in domestic activities:
- Reduction of wastage in irrigation:
- Reduction of wastage in industries:

2. **Improvement of water economy**: These methods/techniques help in recovering the depleting water resources.

- Rainwater harvesting: In rainwater harvesting, the rainwater is collected (harvested) and used for drinking and agriculture.
- Watershed management: In watershed management, a specific land area is used for collection of rainwater or water from rivers or streams.

3**. Development of appropriate technology**:

- Development of technology for treatment of waste water treatment.
- Development of technology to recharge groundwater aquifers.
- Development of suitable technology for desalination of sea water and saline ground water

RAINWATER HARVESTING

Rainwater is fresh and uncontaminated, and if it collected (harvested), it can be used for drinking and agriculture. Rainwater harvesting is a technique of collection of rainwater from the roofs of buildings and storing it in tanks above or below the ground for later use.

This technique is basically designed for water deficit areas (such as: Bundelkhand Region, M.P., Rajasthan etc.) so that they could conserve rain water and utilize it when required. Rainwater is fresh and uncontaminated, and if it collected (harvested), it can be used for drinking and agriculture.

Objectives and advantage of Rainwater harvesting:

To conserve surface run off (=rain water flowing on earth's surface) during monsoon which can be used for domestic and agricultural purposes.

• To recharge ground water aquifers.

• To improve the quality of ground water.

WATERSHED MANAGEMENT

Watershed: A watershed is a specific land area which is managed for collection of rain water or water from rivers/streams or from ground water aquifers and which can be used for drinking, agriculture and other.

Watershed management: In watershed management, a specific land area is used for collection of rainwater or water from rivers or streams.

Objectives:

- Conservation of soil (=land),
- Conservation of water,
- Conservation of vegetation (=plants),
- Recharging of ground water,
- Improving the groundwater table, improving food production (agricultural production) thereby improving the economic conditions of people

Write a short note on floods.

Ans: Covering of extensive (wide) land area with water for several days is termed as flood.

✓ Flood is a natural disaster which results in loss of life and property ✓ Floods occur due to heavy rainfall in a particular region, which causes rivers, streams, canals and lakes to overflow and as a result fields, settlements and roads get submerged for several days in continuation.

In India states like Assam, Bihar and Gangetic parts of Uttar Pradesh are flood prone during rainy season.

The Ganga and Brahmaputra rivers and their tributaries are most susceptible to floods.

Damages caused by floods:

- Loss of life and property
- Fields, crops, settlements (houses) and roads get submerged
- Flood affected areas face acute shortage of food and drinking water.
- Services like transportation, water supplies, telecommunication
- and power are badly hit.

Strategies for flood management (Prevention control and mitigation measures):

- Construction of dams, reservoirs and embankments.
- Diversion of flow of flood waters
- Interlinking of rivers and transfer of water from surplus (excess) to water deficit (shortage) areas.
- Afforestation programme
- Use of advanced technology and communication for flood forecasting and warning
- Public participation through education, training and creating awareness.

Short note on droughts.

Drought is a natural disaster which results in loss of life and property. Drought is defined as a lack or shortage of water for a long period of time in particular region. During drought rainfall is very less.

✓ Drought is a weather hazard. It is indicated by very less rainfall, falling ground water level, drying of wells, rivers and crop failures.

✓ When there is no rainfall, crops do not grow and there is nothing to eat and therefore people start to starving (death due to hunger).

✓ In India some parts of Orissa, Karnataka, Maharashtra, Rajasthan and Gujarat are drought prone.

Causes: Deforestation, Overgrazing, Mining, Rapid Industrialization, Global Warming Etc.

- **Damages caused by drought**:
- Loss Of Life And Property
- "Acute water shortage
- Crop failure
- Loss of biodiversity
- Food and fodder shortage
- Death of Livestock
- Starvation deaths
- Dust Storms
- Strategies for drought management (Prevention control and mitigation measures)
- Plantation or Afforestation programme
- Rain water harvesting

- Watershed management
- Dry farming irrigation techniques like drip irrigation and sprinkles etc should be implemented to irrigate the maximum area with minimum water.
- Growing of drought resistant crops
- Public participation through education, training and creating awareness.

Explain water related diseases. Discuss in detail water borne and water induced diseases.

Water borne diseases are caused by consuming pathogen contaminated drinking water (i.e. the water having disease causing microorganisms).

DISEASE	**CAUSE (PATHOGEN)**	**SYMPTOMS**
Cholera	*Vibrio cholera*	Vomiting and thirst, pain and tiredness, dehydration
Typhoid	*Salmonella typhi*	Prolonged continuous fever, headache, red spots on abdomen and chest
Diarrhoeal	*E.coli*	Dehydration, many times loose motion, weakness
Polio	*Poliovirus*	Permanent paralysis of limbs(legs)
Jaundice	*Hepatitis virus A, B, E*	Tiredness, abdominal pain, fever, weakness

Water induced diseases: Water Induced diseases are caused by consuming chemical contaminated drink Jaundice

DISEASE	CAUSE (CHEMICAL)	SYMPTOMS
Blue-baby syndrome (Methemoglobinemia)	Nitrate in ground water	Blue colour of body of new born babies
Arsenics or Black Foot Disease	Arsenic	Development of black spots on chest and limbs
Fluorosis	Fluoride	Deformity of teeth and bones
Itai-Itai disease	Cadmium (Cd)	Swelling & pain in muscles and joints

Q: **Discuss the Arsenic problem in drinking water.**

The patient develops black spots on chest, back, limbs (hands and legs). The skin becomes hard and fibrous. Severe toxicity can lead to gangrene and skin cancer.

Several cases of Arsenics have been reported from West Bengal and Bihar (India). Bangladesh has some of the most polluted ground water in the world. Around 85% of the area of the country has arsenic contaminated ground water and about 1.2 million people of Bangladesh (Bangladeshis) have been suffering from arsenic poisoning.

SOLUTION FOR ARSENIC (AS) PROBLEM IN DRINKING WATER:

Recent research shows that Pteris vittate (a fern plant) can easily purify water contaminated with arsenic. This method is called Phyto filtration technique

Write a note on fluoride problem in drinking water.

The excess intake of fluoride (higher than the optimum level) in drinking water leads to Fluorosis disease, which results in deformity of teeth and bones. The main source of fluoride in ground water is the rocks which are rich in fluoride.

Fluorosis is an endemic disease. Fluorosis may be-

(1) Dental Fluorosis: Deformity of teeth

(ii) Skeletal Fluorosis: Deformity of bones. Skeletal fluorosis is also called knock-knee disease.

the available data suggest that 19 States in India (including Rajasthan, Andhra Pradesh and Bihar) are endemic for fluorosis (fluoride level in drinking water >1.5 mg/l), and about bout 62 million people in India suffer from dental and skeletal fluorosis.

Q: **Write a short note on Minamata disease.**

Place: Japan

Source: Minamata bay (Japan) in which a plastic factory discharged an estimated 200-600 tons of mercury (Hg) over a period of 36 years. Effect of mercury discharged in to the Minamata bay:

Normally mercury is found in 3 forms: Elemental, Inorganic and Organic Form. Out of these 3 forms, Organic mercury is very toxic in nature.

The plastic factory discharged inorganic mercury in to the bay.

✓ The bacteria at the bottom of the bay converted the inorganic mercury in to organic mercury (Methyl mercury).

The methyl mercury through food chain entered in to the body of fish. When mercury contaminated fish were eaten by human beings it caused mercury poisoning in them, which was called Minamata disease. It was a kind of Mercury Biomagnification.

Result:

The pregnant ladies gave birth to deformed (underdeveloped) babies. Paralysis and other neurological disorders in people. By the year 1992, more than 2209 victims (infected persons) identified and out of these 1403 died.

FOREST RESOURCES (AVAILABILITY), FUNCTIONS OF FOREST, DEFORESTATION-ITS CAUSES, IMPACTS AND CONTROL MEASURES

Functions (or values or importance or role or uses) of forests.

Productive functions	Protective functions	Regulative functions
• We get a lot of valuable products from forests like:	• They provide protection to wild animals	• Regulates atmospheric temperature

• Fruits • Fodder • Fuel wood • Fibres • Wood for furniture • Rubber • Gums • Honey • Medicines etc.	• They provide protection against Soil erosion, drought, flood and landslide. • They help to conserved soil and water.	• Regulates water cycle • Regulates energy cycle • Regulates nutrient cycling (C,H,O,N,P,S cycle) • They purify air (atmosphere) • They control environmental pollution

Deforestation; Deforestation means removal (=cutting down the trees) or destruction of forest cover or felling of tree.

Causes or factors leading to deforestation:

The forests are being cut or cleared because of the following reasons:

For Fuel wood: Trees are felled for getting wood for burning.

For Agricultural purpose: For growing more food to fulfil the need of growing population.

For Development projects: Construction of large dams, hydro- electric plants, railways, roads etc.

Industrial purposes: As raw materials for wood-based industries (paper industry, plywood industry, furniture industry etc.)

Urbanization: For setting up villages, township and cities to accommodate (to provide house) for growing population.

Shifting cultivation: It is also known as jhooming or slash and burn type of cultivation. It is an ancient agriculture method which is very common in north-eastern India. In this type of cultivation farmers destroy the forest by cutting and set it on fire.

ENVIRONMENTAL IMPACTS OF FOREST DEPLETION/DEFORESTATION:

- Loss of greenery
- Difficulty in survival of tribal people.
- Decline in agricultural yields due to decrease in the number of valuable insects like honeybees, butterflies and moths as these insects help in pollination of crops.
- Destruction of habitat of wild animals
- Loss of wild animals
- Loss of biodiversity
- Decrease in rainfall
- Environmental pollution
- Rise in atmospheric temperature
- Change in climatic conditions, Global warming
- Chances of flood, drought and soil erosion and siltation
- Deforestation in Himalayas can cause floods killing people.
- Deforestation in Himalayas can cause floods killing people, damaging crops and destroying homes around the Ganges and its tributaries and the Brahmaputra.
- Reduction in productivity of cropland due to soil erosion
- Ecological imbalance

Control of deforestation:

- By Afforestation & Reforestation programme
- Use of alternative resources of energy in place of fuel wood
- Overgrazing should be checked
- Unauthorized felling of trees should be stopped
- Check on forest fires.
- By Forest Management Programme
- Establishment of National Parks and Wildlife sanctuaries
- Implementation of rule and regulation (For Ex; Forest conservation act 1980)
- By education and awareness.
- Public participation through environmental education, training and creating awareness, for ex: Chipko movement is a living example of people's participation in the conservation of forests,

Forest Conservation movements Tree for a tree" principle should be adopted. Tree plantation movement or Van Mahotsav is being carried out in India since

1950 whereby government and private agencies perform tree plantation programme during July and February every year.

1**. Chipko movement:**

The Chipko (to hug) movement was started in 1973 by local forest dwellers (=local village people) in Garhwal hilly area of Uttarakhand by hugging trees to save them from being cut by commercial wood cutters (as per government order). Gayatri Devi (GAURA DEVI) is known women leader who along with Chandi Prasad Bhatt initiated the movement. Shri Sundar Lal Bahuguna (a Gandhian activist) joined the movement and they become successful in withdrawing (=to take back) the government order for felling (=cutting) trees. In 1980, the Prime Minister (Mrs. Indira Gandhi) imposed (=applied) a 15 year ban on felling (=cutting) trees in Garhwal region. Since then the movement has spread to the other states such as Himachal Pradesh, Karnataka, Bihar and Rajasthan.

2; **Appiko movement**:

It is the Chipko movement of south India. It began in Belegrade village of Karnataka. Some young people (youth) of the village opposed the cutting of teak trees and their complaints were ignored by officials. The youth then decided to launch a campaign and invited Shri Sunder Lal Bahuguna. He gathered the local people to take oath (promise) to protect the trees.

3. **Silent Valley movement**: Silent valley is a peaceful forest area in Kerala. Kerala Electricity Board wanted to construct a dam on river Kuntipujha (flowing through the silent valley forest area) to produce electricity.

An NGO called Kerala Sastra Sahitya Parishad (KSSP) opposed the construction of dam to save the wild life (biodiversity) and environment of the silent valley forest area. Many geologists, scientists and environmentalists supported the NGO and finally they got success and Kerala government had to stop (drop) the project. In 1984, the silent valley forest area was declared as a National Park.

4. **Bishnoi movement** (or Bishnoi culture):

Bishnoi (Bish = twenty, Noi = nine i.e. 20+9=29 Principles)

Bishnois were the followers of Guru Jambheshwar who made 29 principles (=rules). Bishnoi is a traditional community of Rajasthan who made sacrifice for saving the sacred (=holy) khejari tree from being cut by soldiers of the king of Jodhpur (Abhay Pratap Singh) some 300 years ago. Amrita Devi (a Bishnoi woman) sacrificed her life by opposing the soldiers of king Abhay Pratap Singh and did not allow them to cut the trees. Recently the Government of India has

initiated an "Amrita Devi Bishnoi National Award for Wildlife Conservation" in the memory of Amrita Devi.

ENERGY RESOURCES:

Q: **What is energy**?

Energy is the capacity of doing work. Energy resources of two types:

1. RENEWABLE SOURCES OF ENERGY (OR NON-CONVENTIONAL ENERGY SOURCE):

✓ They can be renewed or recycled or regenerated in short period of time.

✓ They are also called non-conventional energy source

✓ They are also called Alternative energy resources.

✓ They are eco-friendly and people friendly energy resources.

For ex: Solar energy, Wind energy, Hydro-energy (hydel, energy), Geothermal energy, Biomass energy, tidal energy.

2. NON-RENEWABLE SOURCES OF ENERGY (OR CONVENTIONAL ENERGY SOURCE):

They are present in limited amount and are exhaustible (= they can be finished).

✓ Once they get finished, they can not be renewed or recycled or regenerated in a short period of time.

✓ They are also called Conventional sources of energy.

✓ They cause environmental pollution.

For ex: Fossil fuels (for ex: Coal, Petroleum, Natural gas) Nuclear fuels (for ex: Uranium, Thorium, Plutonium etc.)

1- CONVENTIONAL (=NON-RENEWABLE) SOURCES OF ENERGY.

FOSSIL FUELS: The fuels which are preserved under the earth's crust as the remains of plants and animals are called fossil fuels. For ex: Coal, petroleum and natural gas

A: **COAL (=BLACK GOLD);**

Coal is a complex mixture of carbon, hydrogen and oxygen.

✓ Small amount of nitrogen and sulphur also occurs in coal.

It is found in deep coal mines under the surface of earth.

✓Coal is mainly burnt in thermal power stations to produce electricity.

Uses (advantages):

- Coal is extensively used in generating electrical energy at thermal power plants.
- Coal is used in manufacture of fuel gases like coal gas.
- Coal is also used as domestic fuel.

Disadvantages (limitations):

Environmental pollution: Burning of coal releases gases such as CO, CO_2, SO2, NO_2 etc, which are a major cause of air pollution. CO_2 is responsible for global warming (greenhouse effect). SO, and NO, are responsible for acid rain.

Coal is non-renewable source of energy (i.e., present in limited amount); it can be finished if it is used continuously.

B: PETROLEUM (CRUDE OIL or ROCK OIL):

petroleum: [petro=rock, oleum=oil

USES (ADVANTAGES):

PETROLEUM PRODUCTS AND THEIR USES	
PRODUCTS	**USES**
Petroleum gas (PG/LPG)	Gaseous fuel
Gasoline or Petrol	Motor fuel, petrol dry cleaning
Kerosene	Household fuel for heating and lighting
Diesel oil	Fuel for diesel engines

Lubricating and oil Vaseline	Lubrication Vaseline, grease, wax for lubrication
Residue coke, asphalt	As a reducing agent, asphalt for making electrodes

Disadvantages (limitations):

Environmental pollution: The various pollutants produced by burning of petrol in automobiles include- carbon monoxide, un-burnt hydrocarbons, oxides of nitrogen (NO, NO_2,) and lead compounds etc.

Carbon monoxide is a poisonous gas which causes respiratory problems in human beings.

NO, is very corrosive and attacks skin. NO_2 is also responsible for acid rain formation.

C: **NATURAL GAS:**

It consists mainly of methane (95%), with small quantities of ethane and propane. Natural gas is formed under the earth by the anaerobic decomposition of vegetable matter lying under water**.**

Uses (advantages):

- It is used as domestic and industrial fuel.
- It is a good fuel because it has a high calorific value.
- It burns with a smokeless flame and cause no pollution

Limitations:

- It is non-renewable.
- The construction of pipelines to carry natural gas over a long distance is quite expensive initially.

PETROLEUM GAS; Petroleum gas is obtained from fractional distillation of crude oil (=petroleum).

It is a mixture of three hydrocarbons: Butane, Propane and Ethane.

For example: Liquefied Petroleum Gas (LPG)

Liquefied Petroleum Gas (LPG); LPG is colourless and odourless (no smell) gas, therefore a strong smelling (=foul smelling) substance called Ethyl Mercaptan (C_2H_5,SH) is added to LPG gas cylinders to help in detection of gas leakage.

Advantage (uses):

It is a good fuel because it has high calorific value.

It is a neat and clean domestic fuel

It burns with smokeless flame and so does not cause pollution.

Disadvantages:

It is easy to handle and convenient to store

It is highly inflammable (i.e., catches fire easily)

Its leakage can lead to explosion.

CNG (Compressed Natural Gas) ; CNG (Compressed Natural Gas) is natural gas stored in cylinders at pressures of 2000 to 3500 pounds per square inch. Natural gas consists mainly of methane (95%), with small quantities of ethane and propane.

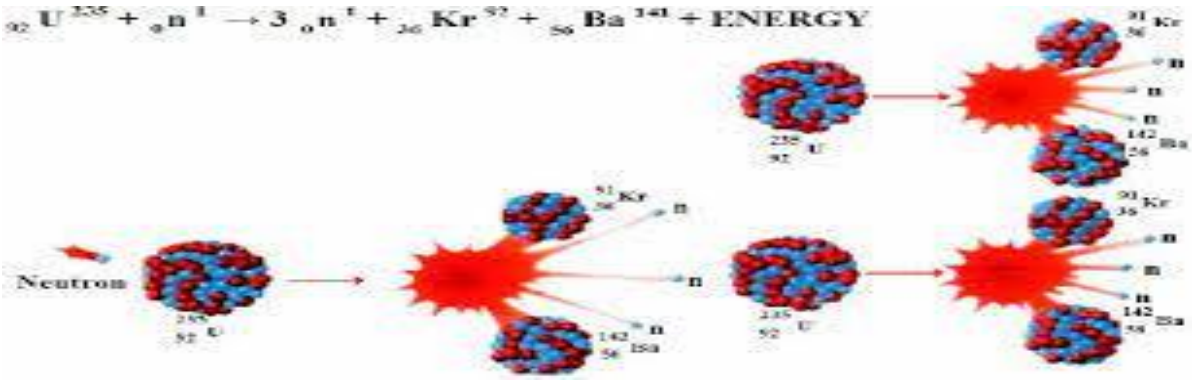
- CNG is being used as an alternative to petrol & diesel in vehicles.
- CNG is the cheapest of all fuels. so it does not cause pollution
- It reduces noise from running vehicles.

Nuclear energy.

Nuclear energy can be generated by 2 types of reactions –

1**. Nuclear fission**:

The process by which a heavy unstable nucleus (such as u-235) is broken down in to two medium weight nuclei by the bombardment of a slow neutron so as to liberate more neutrons and a lot of energy (Nuclear energy).



$_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{36}Kr^{92} + {}_{56}Ba^{142} + 3\,{}_{0}n^{1}$ + ENERGY

2. Nuclear fusion:

It is a nuclear reaction in which two light nuclei (such as isotopes of hydrogen: deuterium) combine to form a heavier nucleus (such as helium) and release enormous energy (nuclear energy).

$_{1}H^{2} + {}_{1}H^{2} \rightarrow {}_{2}He^{2} + {}_{0}n^{1}$ + ENORMOUS ENERGY

NUCLEAR ENERGY IN INDIA: India today has five Nuclear Reactors in operations. Two at Tarapur (Maharashtra), two at Kota (Rajasthan) and one at Kalpakkam (Tamil Nadu) with a combined installed capacity of about 1095 MW representing about 2.5 percent of the total power capacity in the country. Five more units are under construction at Kalpakkam, Narora (Uttar Pradesh), Kakrapara (Gujarat).

Advantages of Nuclear energy:

Small quantity of nuclear fuel produces huge amount of energy o While all the fossil fuels reserves are decreasing, nuclear energy provides a source of limitless energy

Disadvantages of Nuclear energy:

- Nuclear power plants emit radioactive substances which cause radiation pollution.
- People working in nuclear power plants suffer from various health disorders.
- Leakage from nuclear plants can lead to accident (disaster).

2-NON-CONVENTIONAL SOURCES OF ENERGY:

These resources are unlimited or unending, never lasting. For example: Solar energy, wind energy, hydro-energy, ocean energy (tidal energy, sea wave energy) etc.

ALTERNATIVE SOURCES OF ENERGY

The energy sources that can replace t traditional fossil fuel sources (coal, petroleum & natural gas) are called alternative sources of energy.

For ex: Solar Energy, Biomass (Biogas) energy, Wind energy, Wave energy, Tidal energy, Ocean Thermal Conversion Energy, Geothermal energy etc.

SOLAR ENERGY: The energy radiated out by the sun in the form of electromagnetic waves is called solar energy. Sun is almost inexhaustible source of energy.

Harnessing (utilization) of Solar energy:

Solar energy ⟶ **Electrical energy**

[For ex: Solar cell (photovoltaic cell), Solar panel, solar water pump etc.]

Solar energy effect ⟶ **Heat energy by magnifying its**

[For ex: devices like Solar cooker, Solar water heater, passive Solar house, Solar furnace, Solar evaporation (desalination), Solar distillation, etc.

Solar energy $\xrightarrow{\text{Green plants}}$ Chemical energy Or food energy

[For ex**: Photosynthesis process**]

Advantages (merits) of solar energy:

- Inexhaustible
- Pollution free
- Eco-friendly
- People friendly
- Chemical energy or food energy
- Used for generation of electricity in space satellites
- Cost effective

Disadvantages or limitations:

- Solar energy reaches the surface of earth in diffused form.
- Solar energy is not available uniformly all the time and at all the places of the earth.
- Solar energy is stored in Solar cells (=photovoltaic cells) which are very costly.

SOLAR ENERGY BASED DEVICES

A. SOLAR COOKER: Solar cooker is a device which is used to cook food by utilizing the solar energy.

Advantages (merits) of solar cooker:

- It does not require fuel or electricity
- Cost of cooking food is zero as Solar energy is free of cost
- No environmental pollution
- It preserves all the nutrients in the food as food cooks at medium temperature.

Disadvantages (limitations)

- It cannot cook food at night or when sky is cloudy
- It cannot cook food in half an hour. It takes generally 3 to 4 hours to cook food.
- It cannot be used for frying, baking and making chapatis.

B. SOLAR CELLS OR PHOTOVOLTAIC CELLS: A device which directly converts solar energy in to electrical energy is called Solar cell or photovoltaic cell. Solar cells are made from semiconductor elements like silicon and germanium.

Uses (advantages);

- Solar cells are used to generate electricity in man-made space satellites.
- In India Solar cells are being used for providing electricity for street lights at night and for running TV, radio, fans etc.
- Solar panel (group of Solar cells) is used for pumping water
- Solar cells are also used to run watches and calculators

Disadvantages (limitations):

- Solar cells/ Solar panel do not work at night.
- Installation cost of Solar panel is very high.
- A Solar energy installation requires a large area
- Pollution in the environment can also degrade the quality and efficiency of photovoltaic cells.

C- Solar water heater;

D- wind energy.:

The moving air is called wind. The wind has kinetic energy due to its high speed. The kinetic energy of wind is used to run windmills (i.e. machines which work with the kinetic energy of the wind).

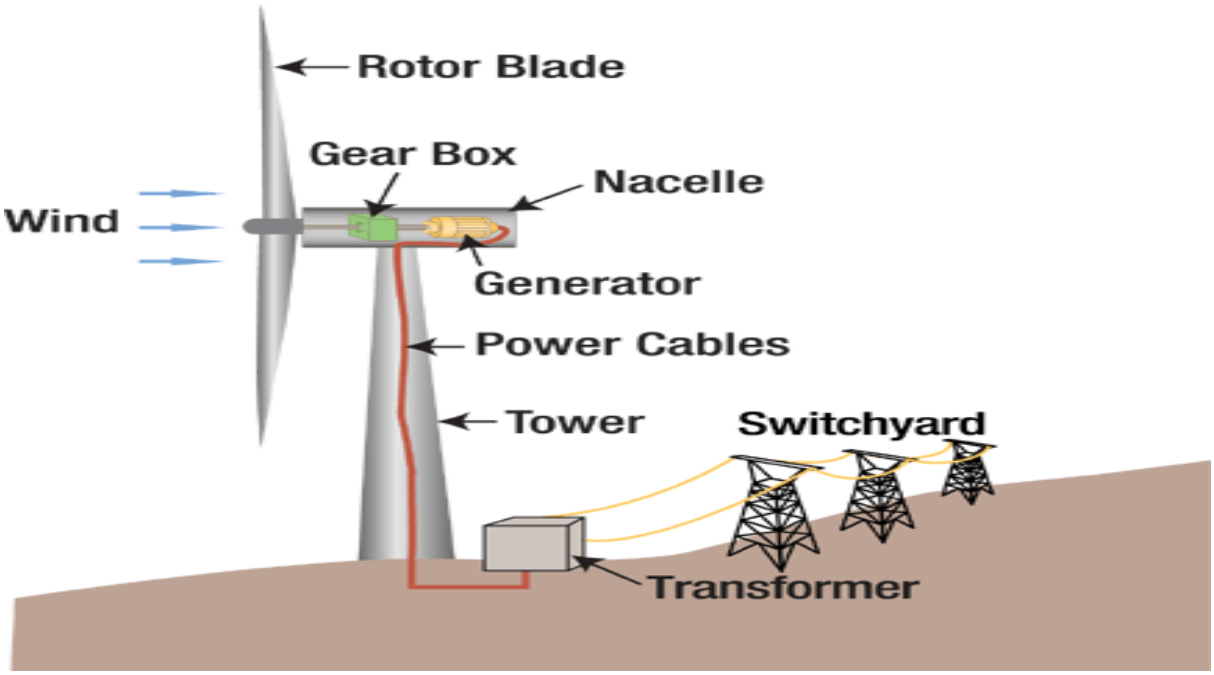
The minimum speed required to operate a wind mill is 15 km per hour (15 km/h).

> The windmills are used for grinding grains, lifting water and to produce electricity.

> The electricity generated by a windmill is small and cannot be used for commercial purpose; therefore, a number of windmills are set up over a large area, known as wind energy farms.

Advantages (merits):-

- Available without any cost
- Non-polluting & eco-friendly.

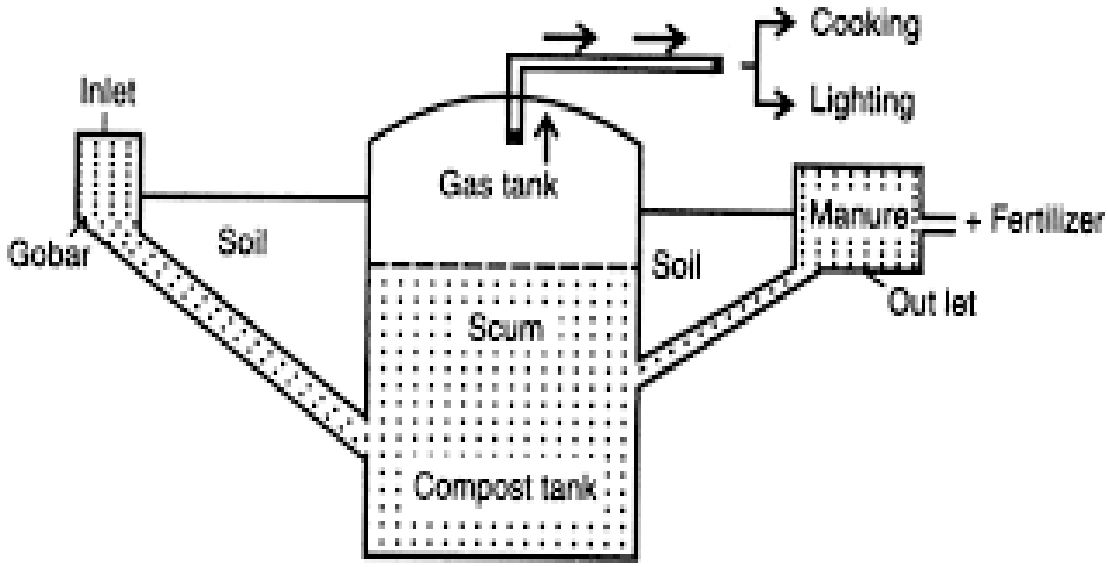


Disadvantages (limitations):

- It is not available all the time and at all the places.
- It depends on the velocity of wind.
- Wind energy farms can't be established everywhere. They can be established only at such places where wind speed is 15 km/h for the most part of the year.
- Setting up of wind energy farms is highly expensive.

BIOENERGY or BIOMASS ENERGY: "Bio-energy refers to various forms of energy generated from biomass by using biotechnological methods". For ex: Biogas,

BIOGAS (GOBAR GAS): It is a fuel obtained from animal dung indirectly. Biogas is produced by anaerobic fermentation (decomposition in absence of air) of animal dung and plant waste in biogas plant (digester tank).

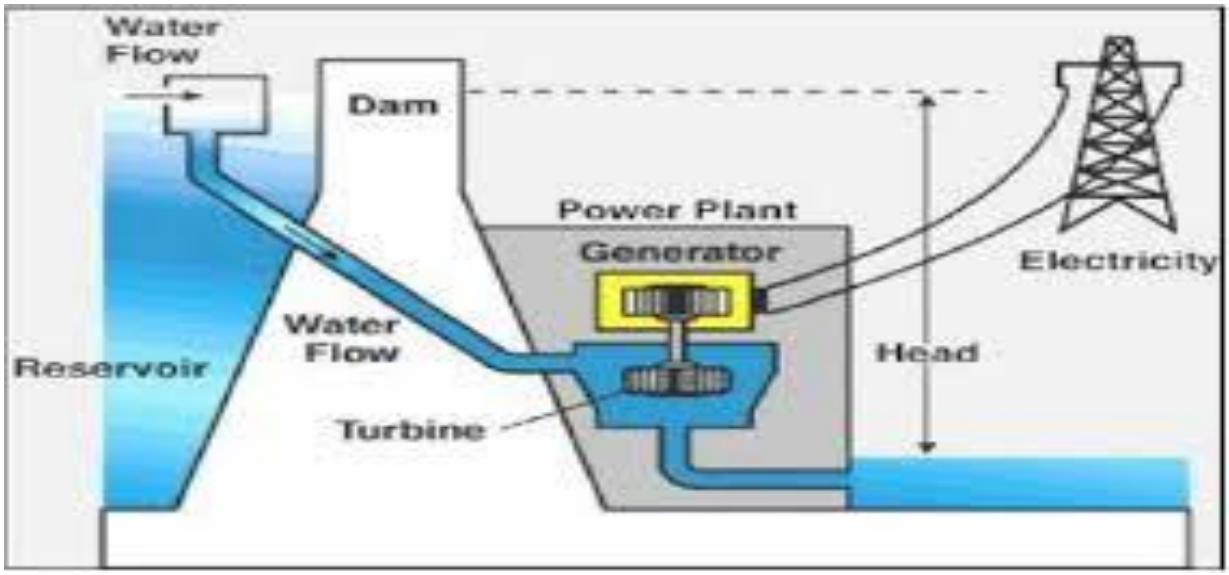


Composition of biogas; Biogas is a mixture of:

Methane (CH4) -63%, Carbon-di-oxide (CO2)-30%, Hydrogen (H.). Hydrogen Sulphide (HS) and Nitrogen (N)-7%

HYDROELECTRIC POWER PLANT OR HYDEL POWER PLANT.

Flowing water has energy (kinetic energy) in it. In a hydroelectric power station, water is stored in a reservoir. This stored water is allowed to fall from a great height which rotates the wheels (blades) of turbine. The turbine rotates the (hydroelectricity). generators to produce electricity



Advantages (merits):

- The water required for generating hydro power is free of cost.
- Hydroelectric power stations are pollution free
- Hydro power is cheapest among the various sources.
- The construction of dam on the rivers helps in flood control and provides water for irrigation and domestic purposes.

Disadvantages (limitations):

- For running a Hydel power station vast amount of water is required.
- Hydro power cannot be stored like other sources of energy. Construction of dam affect the ecology (i.e., deforestation, loss of wildlife).

MATERIAL CYCLE: written your class notes

- **Carbon cycle**
- **Nitrogen cycle**
- **Sulphur cycle**

ENVIRONMENT AND ECOLOGY SYLLABUS
Unit-3:
Pollution and their Effects; Public Health Aspects of Environmental;
Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

ENVIRONMENTAL POLLUTION, ITS TYPES AND TYPES OF POLLUTANTS (ENVIRONMENTAL POLLUTANTS)

The environmental pollution can be defined as: "Any unwanted change in physical, chemical and biological properties of air, water or soil, which adversely affects living beings (various life forms) and property is called environmental pollution."

CAUSE/SOURCE OF ENVIRONMENTAL POLLUTION:

Environmental pollution can be caused either by natural activities or by man-made activities (anthropogenic activities).

(A) **Natural activities or natural causes:** These include-

- Volcanic eruption
- Forest fires
- Soil erosion
- Dust storms
- flood
- pollen grains
- biological decay (decay of plant and animal body)

B) **Man-made activities** :These include-

- Fast growth in human population
- Faster rate of development
- Road and building construction
- Industrialization
- Deforestation
- Urbanization
- Housing
- Mining activities
- Agricultural practices
- Automobile transportation
- Improper disposal of solid waste
- Chemical and nuclear accidents
- Testing of nuclear/atomic bombs
- Nuclear wars etc.

Types of environmental pollution: There are 5 types of pollution.

1. **Air pollution:** "Any unwanted change in physical, chemical and biological properties of air which adversely affects living beings (=various life forms) and property is called air pollution
2. **Water pollution:** "any unwanted change in physical, chemical and biological properties of water which adversely affects living beings (=various life forms) and property is called water pollution.
3. **Soil pollution:** "any unwanted change in physical, chemical and biological properties of soil which adversely affects living being (=various life forms) and property is called soil pollution.
4. **Noise pollution:** unpleasant loud sound is called noise and disturbing level of noise is known as noise pollution.
5. **Radiation pollution:** pollution of air, water and soil with radioactive materials is called radioactive pollution.

Impacts/effects of environmental pollution:
The various impacts of environmental pollution are as follows:
• Global climate change
• Acid rain
• Global warming
• Ozone layer depletion
• Eutrophication of water Bodies
• Biomagnification of chemical pesticides (for ex: DDT) and heavy metals (for ex: Hg, Cd, Pb).
• Various diseases in plants, animals and human beings.
• Air, water and soil borne diseases.
• Acidification of soil and water bodies.
• Corrosion (decay) of property (buildings/monuments)
• Disturbance in aquatic habitat and aquatic food chain
• Loss of wildlife
• Loss of biodiversity
• Ecological imbalance

POLLUTANTS

The substances/agents that pollute the environment are termed as pollutants or environmental pollutants.

TYPES OF POLLUTANTS:
BIODEGRADABLE POLLUTANTS: The pollutants which can be (=decomposed) by microbial action are called biodegradable pollutants For ex: sewage, plant and animal waste.

NON-BIODEGRADABLE POLLUTANTS: The pollutants which cannot be decomposed by microbial action are called non-biodegradable pollutants For ex: glass, plastic, salts of heavy metals, pesticides (such as DDT radioactive materials.)
PRIMARY POLLUTANTS: The pollutants which enter in the environment directly from various sources are called primary pollutants. For ex: (DDT, BHC etc), Unburnt hydrocarbons, NO2, SO2, particulate matter plastic ware etc.
SECONDARY POLLITANTS: These pollutants are formed by reaction be primary pollutants. These are more toxic than primary pollutants For ex: photochemical smog, sulphurous smog, PAN acid rain etc.

CAUSES, IMPACTS AND CONTROL MEASURES OF AIR POLLUTION
Air pollution: "Any unwanted change in physical, chemical and biological properties of air which adversely affects living beings (various life forms) and property is called air pollution.
CAUSES/SOURCE OF AIR POLLUTION OR AIR POLLUTANTS:
NATURAL SOURCES

- Forest fires
- Volcanic eruption
- Dust storms
- Pollen grains from flowers
- Gases from decaying plant and animal waste
- Spores from fungi

MAN MADE SOURCES (ANTHROPOGENIC SOURCES)

- Burning of fossil fuels (coal, petroleum, natural gas)
- Burning of biomass (wood, plant waste, dung cakes)
- Burning of plastic/rubber/ other solid. waste
- Mining activity
- Chemical accidents (for ex: Bhopal Gas Tragedy, Chernobyl Nuclear Accident)
- Dust (SPM) from various industries (ex: cement industry)

From all these sources various air pollutants release into the atmosphere and cause "Air pollution".

AIR POLLTANTS: Air pollutants may be-

1. **PRIMARY POLLUTANTS**: The pollutants which enter in the environment directly from various sources are called primary pollutants. For ex: pesticides (DDT, BHC etc), unburnt hydrocarbons, NO_2, SO_2, CFC's, particulate matter (dust) etc.

2. **SECONDARY POLLUTANTS:** These pollutants are formed by reaction between primary pollutants. These are more toxic than primary pollutants. For ex: photochemical smog, sulphurous smog, PAN (Peroxyl Acetyl Nitrate), O_3, acid rain etc.

EFFECT OF AIR POLLUTION/AIR POLLUTANTS ON ENVIRONMENT:

EFFECT ON ENVIRONMENT/ HUMAN HEALTH

AIR POLLUTANTS	**EFFECT ON ENVIRONMENT/ HUMAN HEALTH**
Carbon-mono-oxide (CO)	It causes blood poisoning.
Carbon-di-oxide (CO_2)	Greenhouse Effect (Global warming)
NO_2	Acid rain, photochemical smog In plants it causes chlorosis (i.e. destruction of chlorophyll contents of leaves)
N_2O	Greenhouse effect
SO_2	Acid rain, sulphurous smog In plants it causes chlorosis and necrosis (i.e. destruction of tissues of leaves)
CFC	Ozone layer depletion, Greenhouse effect
CH_4	Green house effect

Control of air pollution or Abatement of air pollution:

- Environmental education and public awareness
- Through plantation
- Use of alternative fuels of energy like solar energy, wind energy, tidal energy, geothermal energy, biomass energy (biogas or gobar gas) etc.
- **Control of vehicular pollution**:
    - Vehicles should be checked at regular interval.
    - Use of unleaded petrol
    - Use of CNG
    - Use of biofuels (biodiesel/bio petrol)

- Use of filters and catalytic convertors in engines

- **Control of industrial pollution:**
  - Large scale plantation around the factory
  - (Green belt designing). Industries should be installed/setup away from the residence.
  - Factories should have tall chimneys

Rules/acts must be enacted to control air pollution. For ex: Air (Prevention and Control of Pollution) Act, 1981

AIR BORNE DISEASES: An airborne disease is any disease that is caused by pathogens and transmitted through the air. The relevant pathogens may be viruses, bacteria, or fungi, and they may be spread through coughing, sneezing, dust, spraying of liquids etc.

DISEASE	PATHOGEN	SYMPTOMS
Common cold	Virus	Sneezing, sore throat (pain in throat)
Chicken pox	Virus	Rashes on skin, headache and fever
Influenza	Virus	Nausea, headache, dry cough, fever
Mumps	Virus	Swollen throat/salivary gland, headache, fever
Tuberculosis	Bacteria	Cough with or without blood, fever, headache
Meningitis	Bacteria	Chills, fever, headache, coma, death
Whooping cough	Bacteria	Severe coughing, vomiting, fever

Write a short note on coal induced smog or sulphurous smog or London smog or classical smog.

The sulphurous smog is primarily produced by burning of coal or oil at large power plants. When coal or oil is burnt, oxides of sulphur are produced. In atmosphere, the oxides of sulphur react with suspended particulate matter (for ex: dust) under some atmospheric conditions and as a result sulphurous smog is formed. The sulphurous smog formation occurred in London during **December 5-9, 1952**; hence it was named London smog.

Oxides of Sulphur+ Particulate Matter

Atmospheric conditions
stable air cloud cover

↓

Sulphurous smog (grey air)

Effect:

- It caused death of more than 4000 people.
- Health effects were: Bronchitis, Pneumonia and other respiratory diseases.
- In plants it caused leaf necrosis (destruction of leaf tissues)

: **Write a short note on photochemical smog or Los Angeles smog.**

Ans: The photochemical smog is formed when in the atmosphere; oxides of nitrogen react with unburnt hydrocarbons in the presence of sunlight (UV solar radiation).

Oxides of Nitrogen + Unburnt Hydrocarbon

UV Solar Radiation

↓

O_3+PAN+ALDEHYDES+O

PAN=PEROXY ACETYL NITRATE

PHOTOCHEMICAL SMOG

(Yellowish Brown air)

Photochemical smog was first observed in Los Angeles, USA in 1944. hence it was named as Los Angeles smog.

Effects:

- ✓ Hydrocarbons cause cancer
- ✓ Ozone and PAN cause eye watering, irritation of eyes and lungs.
- ✓ Ozone and PAN cause leaf chlorosis (i.e. Destruction of chlorophyll content of leaves) and leaf necrosis (i.e. Destruction of leaf tissues)
- ✓ Aldehydes cause irritation of nose and throat.

CAUSES, IMPACTS AND CONTROL MEASURES OF WATER POLLUTION, WASTE WATER TREATMENT PLANT

Q: **What is water pollution? Discuss its causes (source), effects (impacts) and control measures. Or**

Enumerate various types of pollutants that cause pollution of water bodies. Also discuss how water bodies can be saved from ill effects of uncontrolled discharge of wastewater into them.

Water pollution: "Any unwanted change in physical, chemical and biological properties of water which adversely affects living beings (various life forms) and property is called water pollution.

Causes/sources of water pollution/ water pollutants:

- **Sewage** (i.e., waste water from domestic activities)
- **Effluents** (i.e., waste water from factories/industries)
- **Agricultural waste** (such as chemical fertilizers, chemical pesticides, crop residue etc.)
- Mining activity
- Dumping of solid waste into water bodies
- Radioactive waste (i.e., waste from nuclear power plants)
- Oil spills
- Thermal pollution (i.e., hot waste water form oil refineries, thermal & nuclear power plants)

WATER POLLUTANTS

(A) Physical pollutants: Waste heat (waste hot water) from thermal power plants, nuclear power plants and oil refinery.
(B) Chemical pollutants: may be-
Organic chemicals: for ex: phenols, DDT, BHC, PCB, oil, plastics etc. • Inorganic chemicals: for ex: Nitrates, Phosphates, and Fluorides etc.
Heavy metals: for ex: Hg, Cd, Pb, Zn, Cr, As etc.
(C) Biological pollutants: These include pathogens (disease causing micro-organisms). For ex: Bacteria, Fungi, Algae, Virus, and Protozoa etc.

EFFECT OF WATER POLLUTION/WATER POLLUTANTS ON ENVIRONMENT/HEALTH:

✓ Eutrophication

✓ Biomagnification

✓ Groundwater contamination

✓ Disturbance of aquatic habitat and aquatic food chain Destruction of aquatic life

✓ Loss of wildlife

✓ Loss of biodiversity

WATER BORNE DISEASES; WATER INDUCED DISEASES; same as unit-2

CONTROL OF WATER POLLUTION OR ABATEMENT OF WATER POLLUTION:

i. Sewage and effluents must be treated before discharging them in to water bodies. The sewage and effluent are given three step treatment processes:

• Primary treatment (physical treatment)

• Secondary treatment (biological treatment)

• Tertiary treatment (chemical treatment or advanced treatment)

ii. Dead bodies should not be thrown in to rivers. For ex: Ganga River

iii. Solid waste (for ex: temple waste) should not be dumped in to river water bodies.

iv. Thermal pollution can be reduced by using techniques such as cooling ponds, wet & dry cooling towers.

v. Cattles should not be allowed to take bath in river/lakes and ponds.

vi. The washer man should not be allowed to wash clothes in the river.

vii. Rules/acts must be enacted to control water pollution. For ex: Water (Prevention and Control of Pollution) Act, 1974.

Q: WRITE A SHORT NOTE ON EUTROPHICATION.

EUTROPHICATION:

The surface runoff from agricultural fields carries a heavy load of chemical fertilizers (Nitrates and Phosphates) reach the nearby water body. These nitrates and phosphates act as nutrients for aquatic plants (algae and other unwanted plants). As a result algae and unwanted plants grow in excess, which is known as algal bloom.
Therefore, "Eutrophication is the process of nutrient enrichment of a water body which results in the formation of algal bloom", and such a water body is called eutrophic water body.

EFFECTS OF EUTROPHICATION:

- Algal blooms may totally cover the water surface and release toxic chemicals in water, which kill fish, birds and other aquatic plants.
- Decomposition of algal bloom leads to oxygen depletion in water. Due to lack of dissolved oxygen, water plants and animals die and their dead bodies get deposited at the bottom of the water body. This causes a disturbance in aquatic habitat and aquatic food chain,

Q: WRITE A SHORT NOTE ON BIOMAGNIFICATION OR BIOLOGICAL AMPLIFICATION OR BIOLOGICAL CONCENTRATION
ANS: We use several toxic chemicals like pesticides (like DDT, BHC, DIELDRIN etc.) to protect our crop plants from various diseases. These chemicals are-

- Either washed down into soil and finally enter the underground water table OR
- These chemicals are taken up from the soil by the plants along with water and minerals and thus enter the food chain.

As these chemicals are not degradable, these harmful chemicals get concentrated at each trophic level in the food chain. This phenomenon is termed as biomagnifications or biological magnification.
For ex: In an aquatic ecosystem it was found that while the concentration of harmful chemicals (DDT) in water was only 0.02 ppm, it was 5 ppm in planktons (Phytoplankton's and zooplanktons), 240 ppm in fish that fed on them and 1600 ppm in the birds feeding on these fishes.

POND WATER (0.02 ppm)

⇓

PLANKTONS (5 ppm)

⇓

PLANT-EATING FISH (240 ppm)

⇓

FISH-EATING BIRDS (1600 ppm)

[Biomagnification of DDT in an aquatic food chain, DDT concentration is expressed in ppm)

IMPACT OF DDT: DDT interferes the egg shell formation in many birds so the shells remain thin and break by bird's weight during incubation. As a result, population of many birds have decreased.

PROCESS OF WASTE WATER TREATMENT; Same as your chemistry notes

What are the characteristics of polluted water? Oll, what are the symptoms of water pollution in any water body! ANS:

- Water becomes coloured (greenish/blackish/yellowish)
- Presence of odour (smell) in water,
- Presence of suspended or floating matter in water
- Presence of sediments (soil particles) in water
- Presence of algae and other unwanted plants in water
- Less dissolved oxygen (DO)
- Presence of unwanted chemicals in water (for ex: Nitrates, Phosphates, Pesticides)
- High biochemical oxygen demand (BOD)
- Presence of pathogens (disease causing micro-organism) in water

Q: **What are the characteristics of drinking water or potable water?**

- It should be transparent and colourless.
- It should be odourless (free from smell)
- It should have sufficient Dissolved Oxygen.
- It should not contain any inorganic/organic/biological impurities.

Q: **What is BOD?** (BOD Biological Oxygen Demand): It is the amount of dissolved oxygen required by bacteria in decomposing the organic waste (organic matter) of water,

- ✓ Higher the BOD, lower would be the dissolved oxygen.
- ✓ BODis expressed in mg/Liter.
- ✓ When large amount of sewage is dumped (thrown) in to water, the BOD will increase.
- ✓ If BOD is more than 4000 mg/Liter, it indicates high water pollution (strong sewage in water)
- ✓ Pure drinking water should have BOD less than 1 mg/Liter (1 ppm)

Q: What is COD

(COD Chemical Oxygen Demand): COD test is used to determine the amount of Oxygen needed to oxidize all pollution material. The value of COD is much higher than BOD.

CAUSES, IMPACTS AND CONTROL MEASURES OF THERMAL, SOIL AND NOISE POLLUTION

Q: What is thermal pollution? Discuss its causes (sources), effects (impacts) and control measures.

ANS: THERMAL POLLUTION: "It is defined as the unwanted change in the properties of water due to waste heat in the water (waste hot water) which adversely affect life of aquatic organisms (aquatic plants and animals)".

CAUSES/SOURCES OF THERMAL POLLUTION:

Waste heat or waste hot water is produced from the following industries:

- ✓ Thermal power plants
- ✓ Nuclear power plants
- ✓ Oil refineries etc.

EFFECT OF THERMAL POLLUTION:

- At higher temperature, the solubility of dissolved oxygen of water decreases.
- At higher temperature, the toxicity of chemicals and pesticides present in industrial waste water (effluent) increases, which cause harm to aquatic organisms.
- When waste hot water is discharged in to river/lakes/sea, it adversely affects aquatic plants and animals.
- At higher temperature, the photosynthetic activity of aquatic plants stops.

CONTROL OF THERMAL POLLUTION:

Thermal pollution can be controlled by:

a) **Cooling ponds**: Waste water discharged from industries/factories is stored in ponds where it is cooled by evaporation and then it is discharged in to nearby water body.

b) **Spray ponds**: Waste hot water is sprayed through nozzles in the form of fine droplets and as a result the water gets cooled.

What is soil pollution? Discuss its causes (source), effects (impacts) and control measures, OR

Define land pollution. Discuss the causes of land pollution and its control.

ANS: Soll pollution: "Any unwanted change in physical, chemical and biological properties of soil (land) which adversely affect living beings (various life forms) and property is called soil pollution or landscape pollution.

CAUSES/SOURCES OF SOIL POLLUTION:

Agricultural practices: Use of chemical pesticides (for ex: DDT, BHC, PCB's) and chemical fertilizers (for ex: nitrate and phosphate fertilizers). • Industrial waste: Heavy metals, inorganic and organic compounds as waste product from industries are dumped either on land or into the water bodies.
Radioactive waste: Radioactive waste from nuclear power plants/atomic power plants, nuclear testing and nuclear explosion come in contact with sol and affect soil quality and fertility.
Farm waste: Animal waste and crop waste are dumped in heap on the land, they attract pests and becomes breeding ground for microbes, flies, and rodents (for ex: rat) etc. and thus cause landscape pollution.
Mining activity: Mining activities disturbs the physical, biological and chemical properties of soil

IMPACT OF SOIL POLLUTION/SOIL POLLUTANTS ON ENVIRONMENT:

- The toxic chemicals (=chemical pesticides) and toxic heavy metals like Pb, Cd and Hg present in the soil enter the food chain and get accumulated at each trophic level and affect biological system of living organisms (=animals and human beings).
- Arsenic pesticide can make soil permanently infertile.
- Excessive use of chemical fertilizers adversely affects soil.
- Excessive use of chemical fertilizers makes the soil either too acidic or too alkaline making it unfit for crop growth.

CONTROL OF SOIL POLLUTION

- Excessive use of chemical pesticides and chemical fertilizers should be stopped.
- Chemical fertilizers and chemical pesticides should be replaced by bio-fertilizers and bio-pesticides.
- The farmers should use Blo-fertilizers, Compost, Vermi-compost, Manures and Biopesticides,
- Solid waste should not be disposed in open space. Sanitary land filling method should be adopted for disposal of solid waste.
- Deforestation should be stopped.
- Afforestation and reforestation programme should be promoted.

- Methods like crop rotation, mixed cropping, mixed farming should be Implemented by the farmers.

Q: DEFINE NOISE POLLUTION. DISCUSS ITS CAUSES, IMPACTS AND CONTROL MEASURES.

Noise pollution: "Unpleasant sound is called noise, and disturbing level of noise is called noise pollution". "Noise pollution is measured in decibel (dB).

Central Pollution Control Board (CPCB) has recommended permissible noise levels for different locations:

AREA	**NOISE LEVEL in decibel (dB)**	
	Day	**Night**
Industrial	75	65
Commercial	65	55
Residential	50	40
Silence zone	40	25

Causes/sources of noise pollution:

Natural sources: For ex- thunder and lightening
Man-made sources:

- Industries and machineries
- Road and building construction
- Land and air transportation
- Community activities such as: celebration parties, burning of crackers etc.
- Indoor activities such as: loudly played music systems, televisions, radio etc.

Effect of noise pollution:

- A decibel value above 80dB causes noise pollution.
- Sound becomes uncomfortable above 100 dB.
- A sound more than 115 dB is harmful to ears.
- The first effect of noise is anxiety and stress
- Noise also causes emotional disturbances.
- Noise pollution can cause damage to heart.

Noise pollution causes a rise in blood pressure and blood cholesterol level,

Noise pollution causes stress, nervousness and decrease in working efficiency.

Control of noise pollution:

• Ban on crackers which produce loud sound during festivals.
• Avoid using high sound producing musical instruments.
• Plantation of trees along highways to control noise pollution Plantation of trees around industries (green belt designing). This is called green muffler.
• The noisy industries should be setup away from residence.
• Ear plugs should be provided to workers working in very noisy environment.
• Proper maintenance and lubrication of machineries should be done at regular interval.

SOLID WASTE (INCLUDING E-WASTE), ITS TYPES AND SOLID WASTE MANAGEMENT PRACTICES

Q: What are solid wastes? Discuss their types, effects and name the various methods used to dispose solid waste.

Solid waste: "Any unwanted or rejected or discarded material in solid form which causes environmental pollution is called solid waste.

Types/sources of solid waste:

1-Municipal solid waste or domestic waste

Solid waste (garbage and rubbish) from houses, streets, offices, hotels, market etc. For ex: dust, leaves, cloth, paper, plastic, broken utensils, broken glass, waste food etc.

2-Industrial waste:

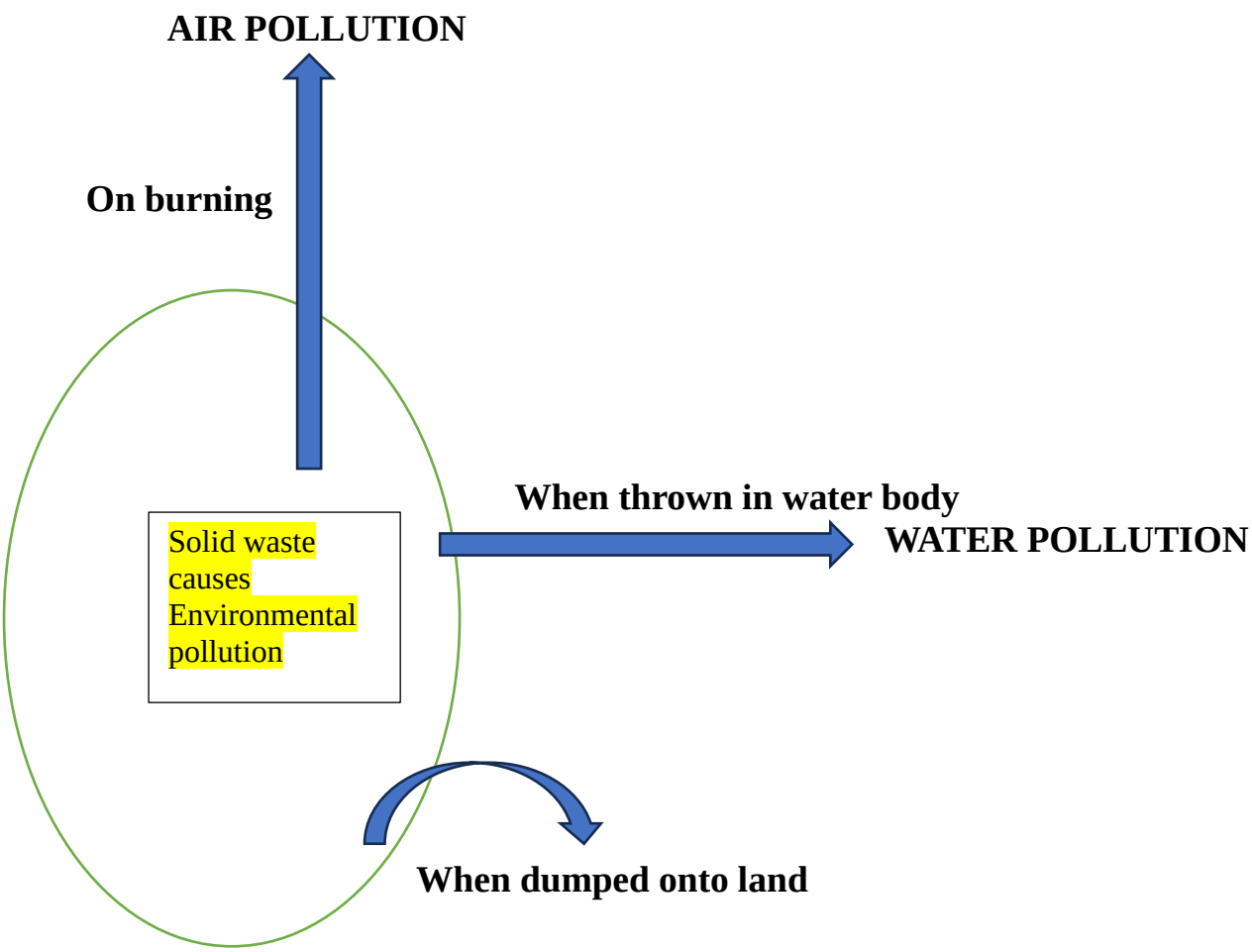
INDUSTRY	**SOLID WASTE**
Packaging industry	Plastic
Cement industry	Cement dust
Mining	Dust
Thermal power plants	Fly ash
Metal industry	Metal scraps

3-Biomedical waste or hospital waste: Solid waste from hospitals, clinics, medical centre and medical institutes. For ex: used syringes, expired medicines, used cotton, x-ray films, glass bottles, plastic bottles, used needles, used bandages etc.

4-Agricultural waste: For ex: crop residue, animal's waste (dung).

5-Electronic waste: Discarded computer parts such as mouse, keyboard, RAM, IC CHIPS, monitor, printer, data cables, wires etc.

IMPACT OF SOLID WASTE ON ENVIRONMENT



- It destroys the beauty of the surrounding.
    - It destroys greenery of the surrounding.
    - It causes land (soil) pollution.
    - "It becomes a habitat of micro-organisms, mosquitoes, flies, rats etc, which spread infectious diseases and affect human health"

SOLID WASTE MANAGEMENT: Solid waste management is a process of collection, disposal, processing and recycling of solid waste in an economic manner with regard to protection of public health.

METHODS OF SOLID WASTE MANAGEMENT:

1**. FOUR "R" PRACTICE: Reduce, Reuse, Recycle and Recovery.**
Reduce: reduction in the use of raw material automatically decreases the production of waste.
Reuse: it is the process of using the same resource again in the same form. For ex: Refilling of cola bottles, reuse of empty glass bottles as containers for storing things •

Recycle: it is the process of using same resource again and again but in a modified form. For ex: Old news papers can be recycled to make tissue paper, card board. Plastic/metallic waste can be recycled to make useful products. Biodegradable waste can be recycled to form compost/manure.
Recovery: it is the process of removal of useful materials from the waste. For ex: metal, energy etc.

2. DISPOSAL OF SOLID WASTE: Some of the commonly used methods for disposal of solid waste are Sanitary Landfilling, Incineration, Pyrolysis, Composting and Vermi-composting.

i. Sanitary landfilling: It is a safe method of disposal of solid waste. In this method solid waste is dumped in on low lying area and is covered with thin layer of soil so that mosquitoes, flies, rats could not make their home/habitat in the solid waste.

Advantages:

- No open burning
- Less odour
- Less cost
- No space problem
- Accommodates a huge amount of waste

Disadvantages:

• Groundwater pollution

ii. Incineration: It is a process of burning of solid waste material at very high temperature to reduce very large volume of waste in to small volume. This method is specially recommended for disposal of biomedical waste.

Advantages:

- Reduced garbage volume
- Occupies less space

Disadvantage:

- More air pollution
- Produces toxic ash

iii. Pyrolysis: It is the process of heating of solid waste in the absence of oxygen.

iv Composting and Vermi-composting:

Composting:

Bio-degradable waste **(dung + plant waste)** —*Micro-organism* (Decomposition)→ Compost **(desi khaad; it is rich in NPK)**

(NPK: Nitrogen , phosphorous and potassium)

Vermi-composting:

Bio-degradable waste (Animal waste+ plant waste) than simple compost) —Earthworm **(digestion)**→ Vermi -compost **(it is richer in npk)**

Q**: Write a short note on E-waste management.**

Ans:
E- waste means any discarded/rejected Electrical or Electronic part/component. For ex: discarded mobile phones, computer, laptops, key-board, mouse, printer, washing machine, TV, radio, refrigerator, data cable, wires, battery etc.

MPACT OF E-WASTE ON HUMAN BEINGS:

POLLUTANT PRESENT IN E-WASTE	**IMPACT ON HUMAN HEALTH**
Phthalates	Affect liver and kidney
Polychlorinated biphenyls	Cancer
Heavy metal (Cadmium)	Affects bone and kidney
Heavy metal (Lead)	Affects nervous system and kidney
Heavy metal (Mercury)	Affects kidney

Unit-4:

Current Environmental Issues of Importance; Global Warming, Green House Effects, Climate Change, Acid Rain, Ozone Layer Formation and Depletion, Population Growth and Automobile pollution, Burning of paddy 'straw

CONCEPT OF GREENHOUSE EFFECT, GLOBAL WARMING AND GLOBAL CLIMATE CHANGE.

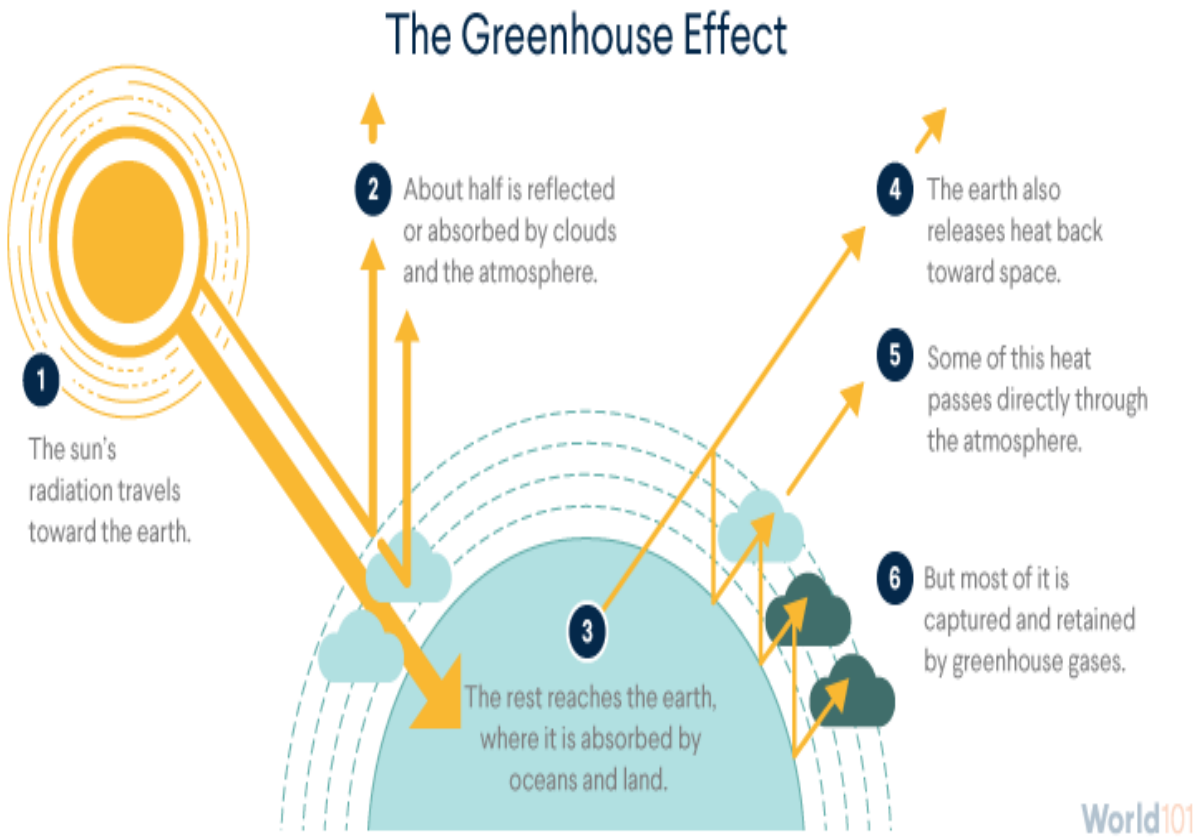
Greenhouse Effect (GHE):

➤ The term Greenhouse effect was coined by **J.B.J. Fourier (1827).**

➤ J.B.J. Fourier stated that earth is a giant (big) greenhouse whose atmosphere (i.e. greenhouse gases such as Carbon-di-oxide, Water vapours, Methane etc.) traps (=capture) radiant heat from sun, warming the planet and giving life to every plant and animal. This phenomenon is known as Natural Greenhouse Effect.

➤ But when the concentration of greenhouse gases increases in the atmosphere, they trap more and more heat radiation, resulting in enhanced (increased) greenhouse effect. Due to enhanced green house effect there is an increase in global mean temperature, which is called "GLOBAL WARMING".

GLOBAL WARMING: "Increase in earth's atmospheric temperature due to increased concentration of Greenhouse gases (primarily CO_2 and water vapours) is known as global warming."



Climate Change or Global Climate Change.

Climate change usually refers to changes of climate caused by human activities such as industrialization, urbanization, deforestation, mining, construction etc. All these activities are responsible for environmental pollution.

Effects of climate change:

- Migration of many animal species towards pole or towards mountain region due to global warming.
- The global temperature increase (rise) will have negative effect (i.e. low production) on food crops for ex; wheat, maize etc.
- The warm marine water will adversely affect the photosynthetic activity of marine plants. Coral reefs will be bleached if water temperature rises above 29.5°C.
- Melting of polar ice/glaciers due to global warming.
- Rise in sea level
- It is estimated that by the year 2080 nearly one third of the total biodiversity
- will go under sea (i.e. will be submerged). Disturbance in rainfall pattern.
- Flood in heavy rainfall areas (i.e., coastal regions) Drought in low rainfall areas
- Loss of wildlife and biodiversity
- Due to global warming more people will be affected by tropical diseases such ask Malaria, Dengue fever ,Yellow fever, Filaria etc..

CAUSES OF GREEN HOUSE EFFECT/SOURCE OF GREEN HOUSE GASES

GREENHOUSE GASES (GHGS)	**% CONTRIBUTION**	**SOURCE**
CO_2	55%	Burning of fossil fuels, burning of biomass (wood), deforestation, respiration of plants and animals
CH_4	15%	Burning and decomposition of biomass, Paddy (rice) cultivation, fossil fuels, Enteric fermentation in cattle's
CFC's (Chlorofluorocarbons)	17%	Air conditioners, refrigerators, aerosol spray

		propellants, sterilizers of medical use
N_2O (Nitrous Oxide)	6%	Burning of fossil fuels, fertilizers, animal wastes
Others	7%	.....

IMPACT OF GLOBAL WARMING/ENHANCED GREEN HOUSE EFFECT:

Climate change: According to scientific estimation global warming may cause the world's temperature to rise by 4.5°c to 5.5°c by the year 2050.

Effects of climate change due to global warming: Migration of many animal species towards pole or towards

mountain region due to global warming. The global temperature increase (rise) will have negative effect (i.e. low production) on food crops for ex: wheat, maize etc.

• The warm marine water will adversely affect the photosynthetic activity of marine plants. Coral reefs will be bleached if water temperature rises above 29.5°C.

Melting of polar ice/glaciers due to global warming.
• Rise in sea level

It is estimated that by the year 2080 nearly one third of the total biodiversity will go under sea (i.e. will be submerged).

• Disturbance in rainfall pattern Flood in heavy rainfall areas (i.e., coastal regions)

• Drought in low rainfall areas (i.e., Central India) Loss of wildlife and biodiversity

• Due to global warming more people will be affected by tropical diseases such as: Malaria, Dengue fever, yellow fever, Filaria etc.

STEPS TO MINIMIZE (I.E. PREVENTION, CONTROL AND REMEDIAL MEASURES) GLOBAL WARMING:

- Environmental education and public awareness
- Plantation (i.e. Afforestation and Reforestation programme)
- Implementation of sustainable agriculture or organic farming.
- Conservation of forests.
- Reduction in use of fossil fuels (coal, petroleum etc.).
- Use of alternative sources of energy. For ex: solar energy, wind energy, geothermal energy, biomass energy etc.

- Use of CNG and biofuels (biodiesel and bio petrol).

Kyoto protocol.

International conference held in Kyoto, Japan during December, 1997 to obtain commitments from different countries for reducing overall greenhouse gas emissions at a level 8% below the 1990 level by 2008- 2012. The developing countries were exempted from mandatory emission control. In 2001 United States rejected the Kyoto protocol as it was too damaging to industries. More recently Russia also expressed his doubts about the Kyoto protocol.

Acid rain: Term acid rain was given by **Robert Angus Smith** in 1852, meaning "the presence of excessive acids in rain water".

In fact Acid rain is a mixture of following acids:

- H_2SO_4
- HNO_2
- HNO_3
- HCI
- $CINO_2$
- $CINO_3$

But the contribution of three acids in Acid rain is mostly in the order:

H_2SO_4 HNO_3> HCI

pH value of normal rainfall: about 5.6

pH value of acid rainfall: less than 5.6

CAUSES OF ACID RAIN FORMATION:

Combustion (Burning) of Fossil Fuels +Burning of Biomass +Volcanic Eruption

↓

Release of Pollutants like Oxides of Sulphur (SO_X), Oxides of Nitrogen (NO_X), CO_2 etc. into the atmosphere.

↓

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SO_2, SO_3, NO_2, N_2O_3, N_2Os, etc. react with water vapours (H_2O) in the atmosphere in the presence of sunlight and as a result strong acidic compounds are formed.

For example: H_2SO_4, HNO_3, HCI etc.

$SO_2 + H_2O \longrightarrow H_2SO_3$

$SO_3 + H_2O \longrightarrow H_2SO_4$

$N_2O_3 + H_2O \longrightarrow 2HNO_2$

$N_2O_5 + H_2O \longrightarrow 2HNO_3$

These Acidic Compounds Get Precipitated in The Form of Rainfall on Earth

Effect on monuments (historical buildings for ex: Taj Mahal): Acid rain causes corrosion of buildings and monuments made up of marble. The famous Taj Mahal of Agra is facing a great danger from acid rain.

The Taj Mahal is gradually losing its marble shine because of the acidic pollutants (SO_2) in air, released from Mathura Oil Refinery which is very close to Taj Mahal,

$CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + H2O + CO_2$

(Marble) (Acid rain) (Calcium sulphate Washed with water)

2. Effect on plants: Acid rain removes the waxy protective coating of leaves, damaging them and inhibits (stop) photosynthesis.

3. Effect on soil: Acid rain increases the acidity of soil due to which beneficial (=useful) microorganisms and earthworms get destroyed.

4. Effect on water bodies (lakes/ponds/rivers etc): Acid rain increases the acidity of lakes, rivers, and ponds which results in death of aquatic organisms (for ex: Phytoplankton's, zooplanktons, and fish etc.).

PREVENTIVE AND CONTROL MEASURES OR HOW TO MINIMIZE ACID RAIN:

- Environmental education and public awareness.
- Plantation (I.e. Afforestation and Reforestation programme)
- Reduce the consumption of fossil fuels for energy production,
- Reduce the consumption of fossil fuels for transportation.
- In place of using. petrol we can use CNG and Biofuels
- Development of more efficient vehicles with improved engines, catalytic converters and filters.
- Installation of scrubbers in factories to absorb acidic gases.
- Use of alternative sources of energy. For ex: solar energy, wind energy, geothermal energy, biomass energy etc.
- Development and use of eco-friendly technology.
- Pollution control laws should be enforced strictly.

Q: What is the importance of ozone layer?

Ans: In the atmosphere ozone is largely found in **stratosphere.** The highest concentration of ozone (around 10 ppm) is found in the stratosphere, approximately 25- 30 km above earth's surface. This is known as the ozone layer. This is also referred as ozone shield because it absorbs ultraviolet solar radiation which is harmful to living organisms. And thus, ozone layer protects the life on earth from severe radiation damage.
The concentration of atmospheric ozone is measured by **Dobson** units (DU). One Dobson unit is equal to a concentration of **1ppb** (parts per billion) ozone.

How is ozone formed in stratosphere?

In stratosphere, ozone continuously forms and dissociates through ozone cycle.
The formation and dissociation of ozone is a continuous process which remains in dynamic equilibrium (balance) and as a result the concentration of ozone is always

Ozone formation:

$$O_2 \xrightarrow{\text{UV rays}} \mathbf{O} + \mathbf{O}$$

(Where O-atomic oxygen)

$$\mathbf{O} + \mathbf{O_2} \xrightarrow{\text{UV rays}} \mathbf{O_3} \quad \textbf{(Ozone)}$$

Ozone dissociation

$$\mathbf{O_3} \xrightarrow{\text{UV rays}} \mathbf{O} + \mathbf{O_2}$$

OZONE LAYER DEPLETION

"Ozone layer depletion refers to the phenomena of reduction in the amount of ozone in the stratosphere".
Scientists (**Farman et. al**.,) in 1985 discovered that ozone layer was thinning (reducing) over **Halley bay in Antarctica (South Pole**) during polar spring (August, September). This thinning of ozone layer was said ozone hole.

ANTHROPOGENIC (=MAN-MADE) CAUSES OF OZONE LAYER DEPLETION

Chlorofluorocarbons, methane and nitrous oxide cause destruction of ozone layer.

- ✓ Chlorofluorocarbons are most damaging to ozone layer.
- ✓ Stratospheric ozone is found to be destroyed by man-made CFCS (Chlorofluorocarbons) for ex: CF_2Cl_2, $CFCI_3$, etc.
- ✓ CF_2Cl_2 and $CFCI_3$, are widely used in refrigerators, air conditioners and as propellant gases.

MECHANISM OF OZONE LAYER DEPLETION:

CF_2Cl_2 and $CFCI_3$, are destroyed by UV radiation in the stratosphere and chlorine is released in this process.

$$CF_2CI_2 \xrightarrow{\text{UV Radiation}} CF_2Cl + Cl^*$$

$$CFCI_3 \xrightarrow{\text{UV Radiation}} CFCl_2 + Cl^*$$

The chlorine radical (Cl*) thus produced is called ozone eater.

$$Cl^* + O_3 \xrightarrow{\text{UV Rays}} ClO^* + O_2$$

(Chlorine monoxide)

$$ClO^* + O_3 \xrightarrow{\text{UV Rays}} ClO_2 + O_2$$

(Chlorine dioxide)

$$O_2 \xrightarrow{\text{UV Rays}} O + O$$

$$O + O_3 \xrightarrow{\text{UV Rays}} 2O_2$$

In stratosphere (ozonosphere) each chlorine atom can destroy thousands of molecules of ozone. This process of destruction of ozone gets speed up (increase) in ice cold temperature.

EFFECT OF OZONE LAYER DEPLETION/OZONE HOLE

The thinning of ozone layer results in more UV-B radiation reaching the earth's surface which adversely affects the life on earth.

1. Effect on human health:

- ✓ UV-B radiation from sun destroys the protein in the body cells and may cause skin cancer, which is known as melanoma.
- ✓ UV-B radiation damages the DNA (Nucleic acid) of the cell and can cause mutation.
- ✓ UV radiation can cause eye damage and cataracts.
- ✓ UV radiation damage functioning of immune system.

2. Effect on plants:

UV solar radiation destroys the chlorophyll content of the leaves and thus affects the photosynthetic activity of green plants.

3. Effect on wildlife (i.e. wild animals and forest):

- ✓ Health hazards to many animal species.
- ✓ Decreased forest productivity

Montreal protocol

Montreal protocol was a landmark international agreement signed on **16 September, 1987** by 34 industrialized countries to protect the stratospheric ozone. Montreal protocol aimed at phasing out ozone depleting substances (ODS) such as chlorofluorocarbons (CFCs) and finding non-toxic alternatives. The CFCs are being replaced by Hydrofluorocarbons (HFCs) and Hydro chlorofluorocarbons (HCFCs), which are potentially less dangerous.

CFCs (Chlorofluorocarbons)

➤ CFCs (Chlorofluorocarbons) are odourless, non-toxic, non-flammable and chemically inert substances made of Carbon, Chlorine and Fluorine.

➤ CFCs (Chlorofluorocarbons) are inert and insoluble in water and therefore they can not be destroyed by chemical reaction or removed by rain. Thus they have a long lasting life in atmosphere.

➤ CFCs (Chlorofluorocarbons) are commonly called Freon gases.

➤ They are used as refrigerants (Freon or CF_2Cl_2), blowing agents and aerosol propellants ($CFCI_1$) for fast freezing of food, industrial sterilization of medical equipment's, metal cleaning & drying and for producing soft polyurethane foams.

➤ CFCs (Chlorofluorocarbons) are strong depletors of stratospheric ozone.

ODS (Ozone Depleting Substances)

- ➢ The substances which can react with ozone present in stratosphere and destroy it are called Ozone Depleting Substances (ODS).
- ➢ The most common ODS are-
- ✓ CFCs (Chlorofluorocarbons) or Freons (for ex: CF_2Cl_2, $CFCI_3$)
- ✓ BFCs (Bromo fluorocarbons) or Halones (for ex: CF_3,Br, CF_2CIBr)
- ✓ Carbon tetrachloride
- ✓ Methyl chloroform
- ✓ Nitrogen oxides
- ➢ CFCs (Chlorofluorocarbons) are strong depletors of stratospheric a Halones are approximately ten times more effective as ozone depletors CFCs.

CAUSES, IMPACTS AND CONTROL MEASURES OF AUTOMOBILE POLLUTION

Automobiles are the largest source of air pollution in cities. They contribute about 60% of air pollution.

COMPOSITION OF AUTOMOBILE EXHAUST:

- Carbon monoxide (77.2%)
- Oxides of Nitrogen (7.7%)
- Hydrocarbons (13.7%)
- Oxides of Sulphur, particulates and some lead compounds

The principal types of automobiles used in our country are Two Wheelers, Three Wheelers, Passenger Cars (Petrol/diesel engine), Trucks and Buses (diesel engine).
The emission from petrol engines are classified as follows:

1. EXHAUST EMISSIONS: These are mostly comprised of CO, NOx (Oxides of nitrogen), particulate matter containing unburnt hydrocarbons.

2. CRANK CASE EMISSION: contains mostly hydrocarbons

3. EVAPORATIVE EMISSION: contains mostly CO and hydrocarbons.

POLLUTANTS	**EFFECT ON ENVIRONMENT/LIVING BEINGS/HUMAN HEALTH**
CO	It causes blood poisoning.
NO_X (Oxides of Nitrogen)	Acid rain, photochemical smog In plants it causes chlorosis (i.e. destruction of chlorophyll contents of leaves). In human beings it causes respiratory irritation, asthma, bronchitis, eye and nasal irritation
UNBURNT HYDROCARBONS (For ex: Benzene, Benzopyrene and Methane)	Unburnt hydrocarbons react with oxides of nitrogen in the atmosphere in presence of sunlight and as a result Photochemical smog is formed. It causes eye irritation and bronchial disorder. Hydrocarbons show carcinogenic effect (i.e., hydrocarbons can cause cancer).
SOx (Oxides of Sulphur)	Acid rain, sulphurous smog In plants it causes chlorosis and necrosis (i.e. destruction of tissues of leaves). In human beings it causes asthma and bronchitis

Particulate matter	Long term exposure to particulate matter in air results in increased rate of bronchitis/asthma and reduced lung function

CONTROL OF AUTOMOBILE POLLUTION:

- ✓ Vehicles should be checked at regular interval.
- ✓ Upgrading of fuel quality and use of cleaner fuel
- ✓ Use of unleaded petrol
- ✓ Adulteration of fuels should also be checked at retail outlets
- ✓ Use of CNG
- ✓ Use of biofuels (biodiesel/bio petrol)
- ✓ Use of filters and catalytic convertors in engines
- ✓ Modification in internal combustion engines to control Co emission.
- ✓ Updating engine design
- ✓ Enforcement of emission norms for new and in-use vehicles.

PARTICULATE MATTERS

: Particulate matters are the discrete mass of any material. For ex: dust, mist, fog, fumes, smoke, fly ash, fur, hair, spores, pollen grains etc. Particulate in atmosphere arise from natural sources like soil and rock debris, volcanic emission, forest fires and anthropogenic activities.

Particulate matter:

Size range: Between 0.0002 µm to 500 µm

Life time: few seconds to several months

Effect of Particulate matter (SPM)

• **PHYSICAL EFFECT:**

- ✓ Reduce scattering of light
- ✓ Reduce visibility of things

• **BIOLOGICAL EFFECT:**

Particulate matter causes respiratory disorders, tuberculosis, lung damage, allergies and many more diseases in animals. According to Central Pollution Control Board (CPCB), particulate matters of 2-4 µm size are mostly deposited in respiratory tract.

PARTICULATE MATTER	**SOURCE**	**HEALTH IMPACTS/RESPIRATORY DISEASES**
Cotton dust	Textile mills	Byssinosis disease in workers in textile mills
Silica dust	Coal mining, silica industry	Silicosis disease in coal mine workers and workers working in silica industry or sand blasting sites
Beryllium dust	Ceramic and metallurgical industries	Berylliosis disease
Coal dust	Coal mining	Black lung disease in coal mine workers. It is also called pneumoconiosis
Manganese dust	Steel industry	Gout disease. The symptoms are swelling of joints of toes, knees and fingers
Asbestos particles	Asbestos industry	Asbestosis (a serious lung disease)

CONTROL MEASURES:

It should be controlled from source i.e. modification of engines can decrease the emission of pollutants.

Gravity settling tanks: These can remove large particles of over 50 µm size. Dirty air is passed through a large tank. Particulates settle down and the cleaner air passes out.

Porous filters: particulate matter over 50 µm size is filtered out by these derives.

Electrostatic precipitators: Electrostatic precipitators can remove over 99% particulate matter present in exhaust from a thermal power plant.

Cyclone collector: The dirty air carrying particulate matter is allowed to rotate, Due to centrifugal force the particulate matter moves towards the boundary of circle. And thus air becomes free from particulates.

What is smog?

➤ Term smog was first used in **London in 1905** for a combination of smoke and fog.

➤ Smog is the combination of smoke and fog.

➤ The name smog was given because the fog in the atmosphere condenses on the carbon particles of smoke to form smog.

➤ Smog reduces availability of light and is harmful to plant life. In plants it causes silvering, glazing and necrosis.

➤ In human beings and animals, it causes respiratory troubles like asthma/bronchitis.

➤ On the basis of composition and method of formation, smog is of two types:

- ✓ Sulphurous smog (or London smog)
- ✓ Photochemical smog (or Los Angles smog)

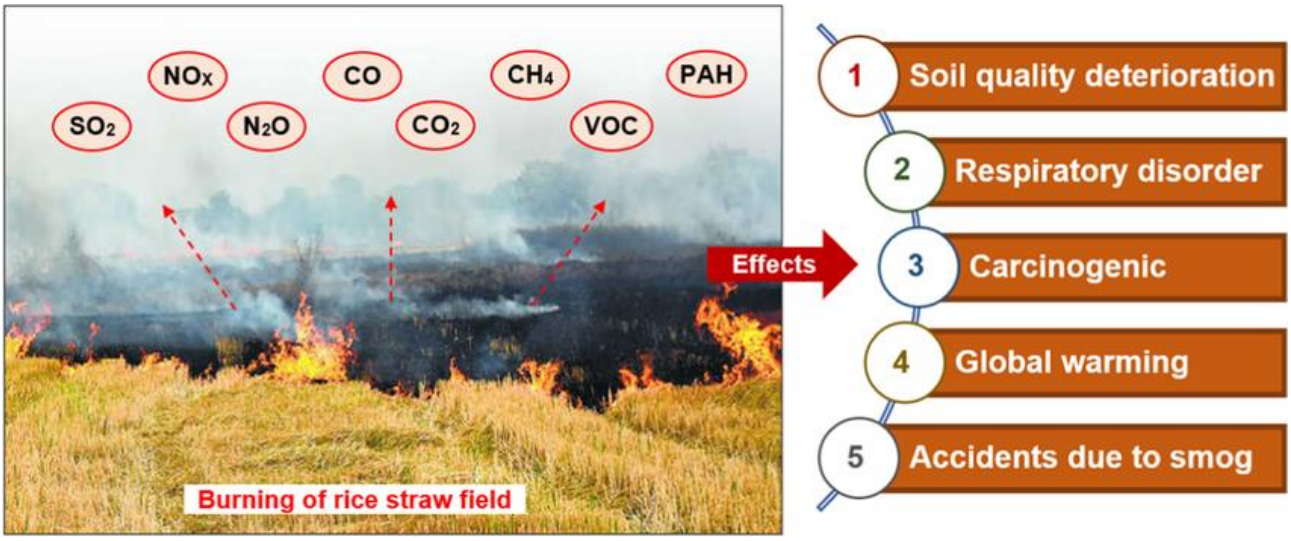
What are aerosols?

➤ Aerosols are chemicals (suspended particulate matter- less than 1 um in size) released in air with force.

➤ An important source of aerosols in stratosphere is jet aeroplanes.

➤ Aerosols contain CFCs (Chlorofluorocarbons). CFCs destroy ozone layer in stratosphere.

PADDY STRAW BURNING OR STUBBLE (PARALI) BURNING



आपके समझने के लिए

पराली (Parali), धान (Rice/Paddy) की फसल कटने के बाद बचे हुए हिस्से को कहते हैं.।

पहले किसान अपने फसल खुद काटते थे तो फसल का बहुत थोड़ा हिस्सा खेतों में रहता थे जिसे जलाने की जरूरत नहीं होती थी, लेकिन पिछले कुछ सालों से धान की फसल की (Paddy Crop) कटाई मशीनों से की जाती है. यह मशीन फसल का सिर्फ उपरी हिस्सी काटती है बाकी का हिस्सा जमीन में ही रह जाता है, जो काफी ज्यादा बचता है. किसानों के पास दूसरी फसल की बुआई करने के लिए कम समय रहता है. ऐसे इन पराली को वो काटने के बजाय जला देते हैं.।
हर साल अक्टूबर और नवम्बर के महीनों में पराली जलाई जाती है ताकि रबी की फसले समय पर बोई जा सके। भारत के कुछ राज्यों जैसे **पंजाब और हरियाणा** में फसल कटने के बाद बचे हुए पराली को जलाते हैं जिससे काफी प्रदूषण फैलता है। उस दौरान पराली से निकले धुएं से स्वास्थ्य संबंधी काफी समस्याएं भी होती हैं। दूसरी तरफ वैज्ञानिकों की माने तो पराली के राख से खेत की मिटटी में पाया जाने वाला राइजोबिया बैक्टीरिया पर खराब असर पड़ता है इस बैक्टीरिया द्वारा ही नाइट्रोजन जमीन तक पहुंचाता है जिससे पैदावार क्षमता बढ़ती है. पराली जलाने से मिट्टी में हुए नुकसान से फसलों की पैदावार पर कम हो जाती है। भारत में खेत में पराली जलाना, भारतीय दंड संहिता की धारा 188 (IPC 188) के तहत गैरकानूनी माना गया है. दोषी पाए जाने पर इस धारा के तहत, 6 महीने का कारावास या 15 हजार रुपए का जुर्माना का हो सकता है

आपके समझने के लिए

RABI CROPS: these are winter crops, grown in winter season (October or November) and are harvested (=cut and collected) in March or April. For example: Wheat, gram, potato, mustard etc.

KHARIF CROPS: these are monsoon crops, grown in (May or June) and are harvested (=cut and collected) in September or October. For example: Rice, Maize, Cotton, jowar, bajra etc.

BURNING OF PADDY STRAW, ITS CAUSES, ENVIRONMENTAL IMPACTS AND CONTROL MEASURES

Paddy straw burning is a process of setting on fire the paddy straw stubble, left after the harvesting of grains of paddy (=rice).
Paddy straw burning is practiced mainly in the **Indo-Gangetic plains of Punjab, Haryana, and UP** to clear the fields for Rabi crops (such as wheat) sowing.

EFFECTS OF PADDY STRAW BURNING:

(A) Environmental Pollution:

➤ Burning of paddy straw emits large amounts of toxic pollutants in the atmosphere which contain particulate matters (such as black carbon (soot)), harmful gases like methane (CH_4), Carbon Monoxide (CO), Volatile Organic compounds (VOC) and carcinogenic Polycyclic Aromatic Hydrocarbons.

➤ These pollutants disperse in the surroundings, may undergo a physical and chemical transformation and eventually adversely affect human health by causing a thick blanket of smog.

➤ These pollutants directly contribute to environmental pollution, and are also responsible for the haze in Delhi and melting of Himalayan glaciers.

(B) Effect on Soil:

Burning husk on the ground destroys the nutrients in the soil.
➤ Burning husk on the ground destroys the nutrients in the soil, making it less fertile.

➤ The heat generated by stubble burning penetrates into the soil, leading to the loss of moisture and useful microbes.

➤ The heat from burning paddy straw penetrates 1 centimetre into the soil, elevating the temperature to 33.8 to 42.2 degree Celsius. This kills the bacterial and fungal populations critical for a fertile soil.

(C) Effect on Human health:

➤ The smog causes irritation in eyes, irritation in nose, irritation in throat, cough and wheezing problem.

SOLUTIONS (REMEDIAL MEASURES) TO THE BURNING PROBLEM:

(A) Agricultural machineries: Farmers can also manage crop residues effectively by employing agricultural machines like:

- ✓ **Happy Seeder** (used for sowing of crop in standing stubble)
- ✓ **Rotavato**r (used for land preparation and incorporation of crop stubble in the soil)
- ✓ **Zero Till Seed Drill** (used for land preparations directly sowing of seeds in the previous crop stubble)

(B) Converting Crop Stubble to Biodegradable Cutlery: Kriya Labs, an IIT- Delhi start-up, has developed a machine that can convert the leftover rice straw into pulp and that is further moulded to produce biodegradable cutlery.

(D) **By converting Crop Stubble into Animal Feed, Manure and Cardboard:**

Eminent Agricultural Scientist M.S. Swaminathan has suggested that the Delhi, Haryana and Uttar Pradesh Governments could set up 'Rice Bio Parks', where farmers could convert stubble into products including paper, cardboard and animal feed.

ANIMAL HUSBANDRY

Animal husbandry is the branch of agriculture that deals with feeding, health, shelter and breeding of livestock. It is an art and science of rearing and breeding of domesticated animals for their commercial and domestic utilization.

Branches of animal husbandry:

- ✓ DAIRY (Cow and Buffalo)
- ✓ POULTRY(Chicken)
- ✓ PISCI-CULTURE (Fish)
- ✓ PIGGERIES (Pig)
- ✓ APICULTURE (Honey bee)
- ✓ SERICULTURE (Silk worm)

IMPACT OF ANIMAL HUSBANDRY ON ENVIRONMENT:

• Animal wastes contain many pathogenic organisms, which may cause diseases in humans and animals.
• Animal waste produces methane, a greenhouse gas, which causes global warming.

• Animal waste causes pollution of water bodies, which kills fish population and other aquatic animals

• Overgrazing destroys the vegetation, which results in soil erosion. Improper treatment of animal wastes results into serious environmental pollution and health hazards. Even Cow dung burning is hazardous to health due to the presence of Benzo(a)Pyrene (BAP) in smoke, which is a cancer causing chemical.

ENVIRONMENT AND ECOLOGY

BAS104

AKTU;B.TECH (FIRST YEAR)

UNIT -5

Environmental Protection; Environmental Protection Act 1986, Initiatives by Non-Governmental Organizations (NGO's), Human Population and the Environment: Population growth, Environmental Education, Women Education

ENVIRONMENTAL (PROTECTION) ACT, 1986

Explain the salient features of environment (protection) act, 1986 or umbrella act.

Or

Why do we refer environment (protection) act, 1986 as an umbrella act? Discuss the environmental protection rules, 1986.

✓ The Environment (Protection) Act is the most important Act in the context of environmental management and was passed by the Parliament in the year 1986.

✓ The genesis of Environment (Protection) Act is in the Article 48A and Article 51A(g) of the Indian Constitution.

✓ This Act empowers the Central Government to prevent, control and abate environmental pollution.

✓ This Act is also called "UMBRELLA ACT" as this Act ensures enforcement of several Acts/regulations concerning Prevention & Control of Environmental Pollution and also Environmental Protection/Safety. The Central Government or any other person duly Authorized is empowered to collect the samples of air, water, soil or other substances as evidence of offences under the Environment (Protection) Act, 1986.

OBJECTIVES:

- Protection and improvement of environment.
- Prevention of hazards of all living creatures and property
- Maintenance of harmonious relationship between humans and their environment

DEFINITIONS (IMPORTANT TERMS USED IN THE ACT):

(a) ENVIRONMENT: environment includes water, air and land and the interrelationship which exists among and between water, air and land and human beings or other living creatures and property.

b) ENVIRONMENTAL POLLUTANT: environmental pollutant means any solid, liquid or gaseous substances which causes pollution.

c) **ENVIRONMENTAL POLLUTION:** means presence in the environment of any environmental pollutant.

d) HAZARDOUS SUBSTANCE: means any substance which can cause harm to human beings, other living creatures and property or environment because of its physical and chemical properties.

SALIENT FEATURES:

- This Act empowers the Central Government to prevent, control and abate environmental pollution.
- It empowers the Central Government to issue directions-
- For the closure, prohibition or regulation of any industry, operation and process.
- For the stoppage or regulation of the supply of water, power or any other service, even without obtaining court orders.
- The Act empowers the Central Government to make rules for the-
    - Standard of quality of air, water and soil for various areas and for various purposes.
    - Maximum permissible limits of concentration for various environmental pollutants (including noise) for different areas.
    - Procedures and safeguards for handling of hazardous substances.
    - Prohibition and restrictions on the location of industries and carrying out processes and operations in different areas.

PUNISHMENT AND PENALTY:

• Any person violating any of the provisions of the Act shall be punished with **IMPRISONMENT UP TO FIVE YEARS OR A FINE UP TO 1 LAKH RUPEES OR BOTH.**

• In case the violation continues beyond a period of one year, the offender/defaulter shall be punishable with the IMPRISONMENT FOR SEVEN YEARS.

ROLE OF GOVERNMENT, NGO'S AND INDIVIDUALS IN ENVIRONMENTAL PROTECTION

Discuss Role of Government and legal aspects in Environmental protection and conservation.

ENVIRONMENTAL PROTECTION, ROLE OF GOVERNMENT & LEGAL ASPECTS:

✓ Environmental legislation is one of the most important means for improving the environment and for controlling pollution.

✓India is the first country in the world to have made provisions for protection and conservation of environment in its constitution.

There is a provision in the constitution of India in favors of protection and improvement of environment. Our constitution reads. "It is the duty of the Government to protect and improvement the environment and to safeguard forests and wildlife of the country." In this regard Government of India has constituted Central and State Pollution Control Board (CPCB/SPCB) and enacted a number of rules and regulations for protecting the environment.

SOME OF THE IMPORTANT ACTS UNDER THESE RULES AND REGULATIONS ARE AS FOLLOWS:

NAME OF ACT/POLICY	**FUNCTION**
The Wildlife (Protection) Act, 1972	Deals with the protection of wildlife in the country.
The Water (Prevention and Control of Pollution) Act, 1974	Deals with the preservation of water quality and control of water pollution.
The Air (Prevention and Control of Pollution) Act, 1981	Deals with the preservation of quality of air and control of air pollution.
The Forest Conservation) Act,1980	Deals with the conservation of forests.
National forest policy 1988	Deals with the conservation of forests.
The Environment Protection act 1986	Deals with the protection of air, water and soil quality and control the environmental pollution.

CENTRAL GOVERMENT

MINISTRY OF ENVIRONMENT AND FOREST [MOEF]

- ✓ CENTRAL POLLUTION -CONTROL BOARD [CPCB]
- ✓ STATE POLLUTION -CONTROL BOARD [SPCB]
- ✓ DEPARTMENT OF -ENVIRONMENT

- ✓ RESEARCH INSTITUTES

Briefly discuss the salient features of Air Act, 1981.

Air Act means: Air (Prevention and Control of Pollution) Act, 1981. This Act was amended in 1987.

This act was passed by the Government in 1981 to regulate and control of air pollution.

OBJECTIVES:

• Prevention, control and abatement of air pollution

• Maintaining the quality of air; and

- Establishment of Boards for the prevention and control of air pollution.

DEFINITIONS (IMPORTANT TERMS USED IN THE ACT):

a) AIR POLLUTION: means "the presence in the atmosphere of any air pollutant."

b) AIR POLLUTANT: means any solid, liquid or gaseous substances

(Including noise) present in the atmosphere which cause air pollution.

c) CONTROL EQUIPMENT: means any equipment or device to control the quality and manner of emission of any air pollutant.

d) BOARD: Board means Central Pollution Control Board (CPCB) and State Pollution Control Board (SPCB).

SALIENT FEATURES:

❖ CONSTRUCTION (ESTABLISHMENT) OF CENTRAL AND STATE POLLUTION CONTROL BOARDS: The Central and State Boards for the prevention and control of air pollution (i.e., CPCB & SPCB) have been set up to monitor the emissions from polluting agents.

a) **FUNCTIONS OF CENTRAL POLLUTION CONTROL BOARD (CPCB):**

• Power to declare air pollution control areas.

• Power to establish standards for emission of air pollutants from automobiles.

• Power to restrict certain industrial plants.

• Power to take/collect sample from polluted areas

• Advises Central Government on matters relating to the prevention and control of air pollution.

PUNISHMENT AND PENALITIES: Defaulter shall be punished with imprisonment of THREE MONTHS or a fine of 10,000 RUPEES or BOTH.

Q: Briefly discuss the salient features of Water Act, 1974.

Water Act means: Water (Prevention and Control of Pollution) Act, 1974. This Act was amended in 1988.

This act was passed by the Government in 1974 to regulate and control of water pollution by industrial, agricultural and municipal effluents.

OBJECTIVES:

- Prevention, control and abatement of water pollution
- Maintaining the quality of water; and
- Establishment of Boards for the prevention and control of water

CONSTRUCTION (ESTABLISHMENT) OF CENTRAL AND STATE POLLUTION CONTROL BOARDS: The Central and State Boards for the prevention and control of water pollution (i.e., CPCB & SPCB) have been set up to monitor the pollution of water bodies.

a) FUNCTIONS OF CENTRAL POLLUTION CONTROL BOARD:

- Power to declare water pollution control areas.
- Power to establish standards for discharge of effluents from industries.
- Power to restrict certain industrial plants.
- Power to take/collect sample from polluted areas
- Power to establish laboratories for analysis of water samples from any stream/river/well or effluent.
- Advises Central Government on matters relating to the prevention and control of water pollution.

b) FUNCTIONS OF STATE POLLUTION CONTROL BOARD:

- Power to declare water pollution control areas.
- Power to establish standards for discharge of effluents from industries.
- Power to restrict certain industrial plants.

- Power to take/collect sample from polluted areas
- Power to establish laboratories for analysis of water samples from any stream/river/well or effluent.
- Advises state Government on matters relating to the prevention and control of water pollution.

PUNISHMENT AND PENALITIES:

Defaulter shall be punished with imprisonment for **THREE MONTHS or a fine of Rs 10,000 or BOTH.**

Q: **Briefly discuss the salient features of Forest (Conservation) Act, 1980.**

FOREST (CONSERVATION) ACT, 1980 (AMENDED in 1988): (Extends to allover India except Jammu and Kashmir)

This Act was enacted in the year 1980, which was later amended in 1988 with the inclusion of more strict penal provisions against violators of the Act.

OBJECTIVES:

• Protection and conservation of forests

• To ensure judicious/proper use of forest products.

DEFINITION:

FOREST: A biotic community, which is composed of trees, shrubs, herbs and grasses.

SALIENT FEATURES:

- Constitution of Advisory committee by Central Government.
- As per this Act, no forest land can be used for no-forest purposes without the prior permission of the Central Government.

PENALTY AND PUNISHMENT:

Any person violating any of the provisions of the Act shall be punished with imprisonment for SIX MONTHS or A FINE OF RS 500 or BOTH.

Q: Briefly discuss the salient features of Wildlife (Protection) Act, 1972.

WILDLIFE (PROTECTION) ACT, 1972 (AMENDED IN 1983, 1986, 1991, 2002):

The Wildlife Act was passed by Indian Parliament in 1972 to protect India's wildlife and their habitat.

DEFINITION:

WILDLIFE: means any animal, bees, butterflies etc. with vegetation.

HABITAT: includes land, water, vegetation which is natural home of any wild animal.

HUNTING: means to capture, kill, injure, destroy or take away & trap away wild animal.

OBJECTIVES AND SALIENT FEATURES:

- To protect the Wildlife and their habitat.
- Prohibition on hunting of specified animals.
- Setting up and management of Protected Areas (such as National Parks, Wildlife Sanctuaries and Biosphere Reserves)

PENALTY AND PUNISHMENT:

• Any person violating any of the provisions of the Act shall be punished with imprisonment **FOR THREE YEARS or WITH A FINE OF RS 25000 or BOTH.**

Q: Explain in brief about MoEF and its functions.

In 1980, Government of India constituted a High Power Committee & then Department of Environment (DOE) was set up. Now, it is a full- fledged Ministry of Environment & Forests (MoEF).

Functions of MoEF: Ministry of Environment & Forests runs Projects related to protection, conservation & management of environment. It performs following functions -

- Survey of Natural Resources - Flora, Fauna & Forests
- Conservation of Natural Resources
- Environmental Impact Assessment (EIA)
- Prevention & Control of pollution.
- Regeneration & Development
- Education on Training & Information
- Legislation & Institutional Support
- International Cooperation & Assistance Programmed.

Q: Discuss the role of Non-Governmental Organizations (NGOs) in creating environmental awareness or public awareness or environmental education or in environmental conservation.

NGOS (Non-Governmental Organizations) play a very important role in environmental education and public awareness. The contribution of NGOs with respect to environment is as follows:

- Environmental education and environmental awareness.
- Public awareness and public participation to conserve environment.
- Conduct environmental movements (such as Chipko Movement, Silent Valley Movement, save seed movement etc).
- Pollution control.
- Wildlife conservation and Biodiversity conservation.
- Afforestation and reforestation programme.
- Also help in bringing awareness and increasing the consciousness level of people regarding good sanitation.
- Creating mass awareness, ensuring public participation in segregation of recyclable material from biodegradable waste at home.

LIST OF SOME NGOS (ENVIRONMENTAL NGOS) IN INDIA:

1. LADAKH ECOLOGICAL DEVELOPMENT GROUP-**LADAKH**
2. DASHOLI GRAM SWARAJ MANDAL (DGSM) –**UTTRAKHNAD**
3. UTTARKHAND SEVA NIDHI PARYAVARAN SHIKSHASANSTHAN (USNPSS)-**UTTRAKHAND**
4. TARA GRAM--**ORCHHA (M.P.)**
5. CLEAN AND GREEN ENVIRONMENTAL SOCIETY (CGES)----**LUCKNOW (U.P)**
6. KALPVRIKSH-**NEW DELHI**
7. WILDLIFE PROTECTION SOCIETY OF INDIA (WPSI)--**NEW DELHI**
8. CENTRE FOR ENVIRONMENTAL STUDIES (CES)- -**NEW DELHI**
9. FRIENDS OF RIVER NARMADA--**MADHYA PRADESH**
10. GREEN FUTURE FOUNDATION- **MAHARASHTRA**
11. SATPUDA FOUNDATION---- **MAHARASHTRA**
12. KERALA SASTRA SAHITYA PARISHAD (KSSP)-- **KERALA**

Discuss the role of an individual in prevention of pollution.

Or

Discuss the role of an individual in protection of environment or natural resources.

Following are some of the important guidelines, which an individual can pursue (follow) to protect the environment/natural resources and in prevention of pollution.

- Try to live in such a way, which is close to nature.
- Adopt sustainable lifestyle.

- Try to use only eco-friendly products.
- Plant trees and take care of them.
- Minimize the use of substances that cause pollution.
- Minimize the use of chemical fertilizers and pesticides. Emphasize more on bio-fertilizers and bio-pesticides.
- Try to minimize the use of fossil fuels as far as possible.
- Try to reduce, reuse and recycle the waste.
- Save water in washing, flushing and leakage.
- Save energy by consuming it economically and avoid wastage.
- Reduce the use of plastic bags.
- Stop smoking as smoke pollutes the home, offices and surroundings of a smoker.

POPULATION: CHARACTERISTICS, POPULATION ASPECTS POPULATION & ITS GROWTH AND RELATED

Q: Define the term population. Discuss its types and various characteristics or attributes.

In ecology, a population may be defined as a group of organisms of the same species occupying a particular space (i.e. area). In simple terms, population of a country or any area is the total number of people who live in it. The scientific study of population is called **DEMOGRAPHY**.

CHARACTERISTICS OR ATTRIBUTES OF POPULATION: Population has specific characteristics that allow it to be distinguished from one another.
For example-
Population size (N): It generally expressed as number of individuals in a population.

Population density: It is defined as number of individuals per unit area.

Birth rate (or Natality rate): It refers to the rate at which new born babies are given birth.

✓ Birth rate (B) is calculated by the number of individuals born (=live births) per 1000 individuals of the population per year.

✓ It is expressed by: B=Nn/t (where B birth rate, Nn=number of new borns, t=time)

✓ **Total Fertility Rate (TFR):** It is calculated by the average number of children that a women bears in her lifetime. The value of TFR is low (=2) in developed countries and high (= more than 4) in developing countries. It is one of the main parameter to determine the population growth of a Nation.

Death rate (Mortality rate): It refers to the rate at which Individuals die either naturally or unnaturally.

✓ Death rate (d) is calculated by the number of individuals who die per 1000 individuals of the: Death rate (d) is calculated by the number of individuals who die per 1000 individuals of the population per year.

✓ It is expressed by: d=D/t (where d= death rate, D=total no. of deaths, t= time)

✓ **INFANT MORTALITY RATE (IMR):** It is the percentage of infants that die out of those born per year. It affects the future growth of population of a country. It is high in developing countries and very low in developed counties.

Growth rate: It is the difference between birth rate and death rate,
If birth rate exceeds the death rate, the population grows.
If birth rate is lower than the death rate, the population declines.

Sex ratio: it refers to the number of males relative to the number of females in the population. It is also known as male-female ratio

Age structure or Age distribution: It is the number or percentage of individuals in a population in different age groups.

Population equilibrium: The condition when birth rate and mortality rate become equal.

Population dispersal: It is the movement of individuals into or out of the population area. It takes three forms.

- **Emigration:** one-way outward movement
- **Immigration**: one-way inward movement
- **Migration**: periodic departure and return

• **Population fluctuations**: Once a population becomes established in a certain area and has reached the equilibrium, its number tends to vary above and below the equilibrium from time to time. This phenomenon is called population fluctuation.

Q: Write a short note on population growth.

Population growth refers to the net increase in the population level of any country over a period of time.

The change in population size is given by the equation-

$$Nt = N_0 + B + I - D - E$$

Where-

N_0= Initial size of population

B= Birth rate

I= Immigration rate

D= Death rate

E= Emigration rate

- ✓ If Nt=0, it means there is no population growth
- ✓ If more individuals are added to the population than are lost, the population will show positive growth.
- ✓ If more individuals are lost than added, the population will show negative growth.

CHARACTERISTICS OF POPULATION GROWTH:

a) Exponential growth: This refers to the growth of population in geometric progression (i.e. 1, 2, 4, 8, 16...)
b) Doubling time: The time needed for a population to double its size at a constant annual rate is known as doubling time.
C) Age structure: In most of population, individuals are of different age, and the proportion of individuals in each age group is called age structure or age distribution. Age structure can be represented with the help of Age pyramid or Age-sex pyramid. Age pyramids are formed on the basis of people belonging to different age groups such as:

• Pre-reproductive (0-14 years)

• Reproductive (15-44 years)

• Post-reproductive (45 years and above)

Discuss population explosion in Indian context. Discuss the factors responsible for population explosion in India. What should be the objectives of a sound population policy?
Or
Discuss some steps to reduce pressure caused by population explosion.

POPULATION EXPLOSION:

The sudden and rapid increase of human population which crosses the growth rate of food grain in an economy is termed population explosion.
Population explosion has become a serious world concern.

FACTORS RESPONSIBLE FOR POPULATION EXPLOSION IN INDIA

- ✓ Illiteracy (Most of the rural population are still illiterate, ignorant and superstitious)
- ✓ Traditional beliefs (like children are regarded as gift of God and a sign of prosperity) I High mortality rate due to better medical facilities
- ✓ Economic reasons (like children are considered to be helping hands to increase the family income)
- ✓ Religious and social customs
- ✓ Desire for a male child (as male child is essential for keeping up the name of the family and male child is usually a great help to the aged parents) Lack of recreation other than sex due to poor standard of living and poverty.

HARMFUL EFFECTS (CONSEQUENCES) OF POPULATION EXPLOSION:

The population explosion will result in misery (=unhappiness), poor health, increase of urban slums and decrease in per capita income (i.e., economic problems) due to following reasons:

- Shortage of natural resources like land, minerals, wood, fuel etc.
- Energy crisis
- Shortage of raw materials, housing and lack of educational facilities.
- Acute shortage of clothing and shelter.
- Unemployment

• Problem of sewage disposal

• Pollution of air, water and land.

• Danger of epidemic diseases.

• Eco degradation

STEPS TO OVERCOME THE PROBLEM OF POPULATION EXPLOSION:

• The concept of value education should be implemented.

• People should be made aware of the advantages of having small families and at the same time, the disadvantages of having many children.

• Shedding away the old faiths and beliefs that more the children, greater would be the earning capacity of the family.

• People should be advised to adopt family planning methods by which they can prevent pregnancy (contraception).

• Adoption of woman welfare programmes

The inverted red triangle has become sign in India for Family Welfare. It is prominently displayed at all such offices and hospitals where any help or advice about family planning is available free of cost.

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• There should be Legal Enactments for the people with more than a specified number of children.

ENVIRONMENTAL EDUCATION AND NEED FOR WOMEN EDUCATION

Q: What is environmental education? Discuss its objectives, principles and types.
OR
What should be the guidelines for excellence for environmental education?
OR

What is environmental education? How it can protect the environment? Explain the need of education (environmental education) for controlling the pollution.
OR
Why the education and awareness in common people is necessary for pollution control?
OR
Environmental education can play an important role in environmental protection. Justify

Environmental education is education through, about and for environment. Environmental education means the educational process dealing with man's relationship with his natural and man-made surroundings (environment).

OBJECTIVES OF ENVIRONMENTAL EDUCATION:

- ✓ To promote and spread environmental awareness among all sections of the society.
- ✓ To promote environment education educational/scientific/research institutions. through existing
- ✓ To ensure training and manpower development for environment education, awareness and training.
- ✓ To encourage Non-Governmental Organizations, mass media and other concerned organizations for promoting awareness about environmental issues among the people at all levels.
- ✓ To use different media including films, audio, visual and print, theatre, drama, advertisements, hoarding, posters, seminars, workshops, competitions, meetings etc. for spreading messages concerning environment and awareness.
- ✓ To mobilize people's participation for preservation and conservation of environment.

PRINCIPLES OF ENVIRONMENTAL EDUCATION

Environmental Education must involve everyone.

- ✓ Environmental Education must be lifelong

- ✓ Environmental Education must be holistic.
- ✓ Environmental Education must be practical.
- ✓ Environmental Education must be in harmony with social and economic goals and accorded equal priority.
- ✓ Environmental Education must help learners to discover the real cause of environmental problems.
- ✓ Environmental Education must emphasize the complexity of environmental problems so as to develop critical thinking and problem solving skills in learners.

TYPES OF ENVIRONMENTAL EDUCATION:

Environmental education may be-

A. FORMAL ENVIRONMENTAL EDUCATION: Formal environmental education is provided through Schools, Colleges and Universities as a part of their curriculum.

B. NON-FORMAL ENVIRONMENTAL EDUCATION:
It is provided through NGOs, Eco-clubs and Mass media through campaigns, rallies, padyatras, public meetings, workshops, training programmes, exhibitions, essay/debate/painting/poster competitions, folk dances & songs, street theatre, puppet shows, preparation and distribution of environmental education resource materials etc.

Q: Discuss women education and strategies adopted by Government of India related with it.
OR
Discuss the strategies and policies adopted by Government of India for development of women education. Describe briefly the various schemes launched for women education in India.
OR
Discuss how women education aid with development of India.
OR
Discuss women education and strategies and policies adopted by Government of India related with it.

WOMEN EDUCATION IN INDIA:

Women in India constitute 50 per cent of the country's human resources and their contributions are vital for the Nation's progress. But the magnitude of illiteracy among women is very high. Only **65.46** per cent of women are literates as per **2011** census. Even though the education system expanded very rapidly, the gender gap in literacy remains conspicuous by its presence. The Constitution of India confers on women, equal rights and opportunities in all fields. The Government of India has endorsed the same through its Plans, Policies and Programmes. In spite of all these, women in India have not been able to take full advantage of their rights and opportunities in practice for various reasons

THE FACTORS/OBSTACLES RESPONSIBLE FOR LOW FEMALE LITERACY RATE IN INDIA:

- ✓ Gender based inequality.
- ✓ discrimination and economic exploitation
- ✓ Occupation of girl child in domestic chores
- ✓ Low enrolment of girls in schools
- ✓ Absence of female teachers in schools.
- ✓ Schools established in faraway places etc.

STRATEGIES ADOPTED BY THE GOVERNMENT FOR INCREASING FEMALE LITERACY IN THE COUNTRY:

The main strategies adopted by the Government for increasing female literacy in the country include:

1. National Literacy Mission (1988) for imparting Functional Literacy.
2. Universalization of Elementary Education.
3. Non-Formal Education.

<u>In addition to the above, the following are the some strategies recommended to encourage girls/women's education in India:</u>

- Providing free and compulsory education for all girls up to 20 years.
- Establishment of more number of primary school.
- Bringing out changes in the attitudes of parents and in the society.
- Removal of gender bias in the school curriculum
- Establishment of more number of NFE (Non-Formal Education)
- schools to enroll girls and women of different age groups.
- Widening the scope of Distance Education Programme to cover all categories of people especially rural women.

VARIOUS SCHEMES LAUNCHED BY GOVERNMENT ARE AS FOLLOWS:

(A) FOR WOMEN EMPOWERMENTOR WELFARE:

a. **INDIRA MAHILA YOJNA:** This is a scheme for women development and is carried out through the Anganwadis programme.
b. **MAHILA SAMRIDHI YOJNA:** This is a scheme to create the habit of saving money among the rural women. It has been merged into Indira Mahila Yojna.
c. **BALIKA SAMRIDHI YOJNA:** This is a scheme to enroll the girl child in the schools and provision of scholarship for education of the girl child when she attends the school.
d. **SWASHAKTI**: This is a scheme to improve the conditions of poor rural women.
e. **SWADHAR**: This is a scheme to provide assistance for food, shelter, health, training and rehabitation.
f. **SWAYAMSIDHA**: This is a scheme to help the women groups to make them self-reliant.